

McKnight Road Retail Development



Traffic Impact Study

Prepared September 18, 2026 · 40.5250, -80.0130

Simple Impact Studies — sample draft

Traffic Impact Study

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1.0 EXECUTIVE SUMMARY

This Traffic Impact Study (TIS) evaluates the transportation impacts of the proposed McKnight Road Retail Development located within Pittsburgh MSA, Pennsylvania. Analysis follows PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual (2019) and applicable PennDOT standards. The study area encompasses 36 intersections within a 2.00-mile radius from the site, analyzed for Land Use 820 (Shopping Center / Retail) with a development size of 200 1,000 sqft GLA.

Key Findings:

- 6 intersections are projected to drop one or more LOS grades under Build conditions.
- 9 intersections operate at LOS E or F under Build conditions — mitigation analysis is required per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual.

36 INTERSECTIONS	6 LOS DROPS	9 AT LOS E/F	48.4s WORST Δ DELAY
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2.0 PROJECT DESCRIPTION

Project name	McKnight Road Retail Development
Land use	820 — Shopping Center / Retail
Development size	200 1,000 sqft GLA
Site coordinates	40.5250°N, 80.0130°W
Jurisdiction	Pittsburgh MSA
State	Pennsylvania
State DOT	Pennsylvania Department of Transportation (PennDOT)
Opening year	2028
Study type	Traffic Impact Study (TIS)
Controlling document	PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual

Site plan, surrounding land use, and access configuration are dependent on the final site plan and are not auto-generated by this tool. Final submittal should incorporate a site plan figure per controlling guidelines.

3.0 METHODOLOGY

3.1 Governing Documents and Controlling Agency

This study is prepared in conformance with PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual (2019). The reviewing agency is the Pennsylvania Department of Transportation (PennDOT).

3.2 Pre-Application Methodology Meeting

A pre-application methodology meeting with PennDOT is required before data collection. PennDOT Pub 282 requires a scoping meeting with the PennDOT District Traffic Unit before data collection. A written scope confirmation must be received before commencing counts.

3.3 Study Area and Analysis Periods

Study area: all signalized and unsignalized intersections and roadway segments within 2.00 miles of the site that receive ≥10% of project traffic. Primary analysis periods: weekday AM peak hour and weekday PM peak hour. Additional periods (Saturday peak, midday) are required only where special characteristics warrant, consistent with PennDOT Publication 282 — Guidelines for the Development

and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual.

3.4 Analysis Scenarios

Three scenarios evaluated at each study location:

- Scenario 1 — Existing Conditions: field-collected counts for current peak-hour volumes and geometric conditions.
- Scenario 2 — Future No-Build (2028): background growth applied at 1.50%/yr compound over 2 years, without project traffic.
- Scenario 3 — Future Build (2028): No-Build volumes plus distributed project-generated trips.

3.5 Trip Generation

Trip generation is calculated using Public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), as used in Pennsylvania screening practice. PennDOT Pub 282 explicitly references ITE. The published rate or equation for Land Use Code 820 (Shopping Center / Retail) is applied to a development size of 200 1,000 sqft GLA. Pass-by credit of 30% and internal capture of 0% are applied per standard screening procedures and agreed in the methodology meeting.

3.6 Background Growth Rate

Background traffic growth: 1.50% per year (PennDOT traffic count data (PENNDOT Count Data — online count station database); DVRPC (Philadelphia metro), SPC (Pittsburgh metro), LVPC (Lehigh Valley), or applicable MPO travel demand model.). Source: Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer..

3.7 Level of Service Standards

Acceptable LOS per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual: LOS D in urbanized areas; LOS C in rural areas. LOS D in urbanized areas; LOS C in rural areas per PennDOT Pub 282. Pennsylvania has one of the most detailed published TIS guidance documents in the US (Pub 282) — follow it closely. The controlling standard evaluates LOS per the Highway Capacity Manual (current edition) via HCS and/or Synchro; this screening applies the openly-published Webster/Akçelik signalized model, superseded by those tools at submittal. Synchro/SimTraffic and HCS standard per PennDOT Pub 282. PennDOT District 6 (Philadelphia) may require VISSIM for signalized corridor studies.

4.0 EXISTING CONDITIONS

4.1 Roadway Network

The existing roadway network within the study area was inventoried to document functional classification, number of travel lanes, posted speed limits, traffic control, and access management characteristics. Data sources: PennDOT functional classification maps, field observations, and Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer..

4.2 Traffic Volumes

Traffic counts were collected during weekday AM (7–9 AM) and PM (4–6 PM) peak periods. Count data represents typical weekday conditions, adjusted to peak-season or peak-annual-day volumes per PennDOT count methodology. AADT base: — vehicles per day (2028).

Base AADT	— vpd
Growth rate applied	1.50%/yr

Growth source

Explicit override — 1.50%/yr was supplied with the request and applied in place of the engine's own rate. No measured growth rate is wired for this region, so there is no measured value to compare the override against. The applied rate is not derived from measured data; its basis must be stated by the preparer.

Count collection period

Weekday AM + PM peak (field-collected)

4.3 Existing Intersection Operations

Intersection	Control	Dist (mi)	Exist LOS	Delay (s/veh)
Near McKnight Road	Signal	0.38	C	30.3
Babcock Boulevard & Rochester Road	Signal	0.40	D	36.1
Siebert Road & McKnight Road	Signal	0.41	B	13.1
McKnight Road & Braunlich Drive	Signal	0.44	A	9.2
Babcock Boulevard & Siebert Road	Signal	0.44	D	36.1
McKnight Road & Hilands Place	Signal	0.58	C	21.1
Cemetery Lane & Babcock Boulevard	Signal	0.66	D	41.9
Glenmore Avenue & Rochester Road	Signal	0.71	C	25.1
Babcock Boulevard & Evergreen Road	Signal	0.85	B	11.0
McKnight Road & Patrick Place	Signal	0.87	A	9.1
Perry Highway & West View Park Drive	Signal	1.07	A	7.5
Center Avenue & Perry Highway	Signal	1.08	B	16.1
Perry Highway & Chalfonte Avenue	Signal	1.12	C	24.8
Perry Highway & Highland Avenue	Signal	1.14	F	96.0
McKnight Road & Ross Park Mall Drive	Signal	1.17	A	9.1
Bellevue Road & Perry Highway	Signal	1.18	D	37.0
Perry Highway & Rochester Road	Signal	1.19	B	10.8
McKnight Road & McKnight Circle	Signal	1.19	A	9.1
Perry Highway & Perry Lane	Signal	1.21	C	24.8
Center Avenue & West View Park Drive	Signal	1.26	A	5.1
McKnight Road & Nelson Run Road	Signal	1.29	A	4.8
Babcock Boulevard & Thompson Run Road	Signal	1.39	D	37.8
McKnight Road & Johnanna Drive	Signal	1.45	B	11.5
Perrysville Avenue & Perrysville Road	Signal	1.46	B	15.9
Perry Highway & Good Lane	Signal	1.56	C	21.1
Perrysville Avenue & Cemetery Lane	Signal	1.62	F	264.6
Perry Highway & Lindley Lane	Signal	1.63	D	39.0
Three Degree Road & Babcock Boulevard	Signal	1.68	F	116.8
Peoples Road & Babcock Boulevard	Signal	1.76	F	133.4
Perrysville Avenue & West View Avenue	Signal	1.78	F	219.8
McKnight Road & McIntyre Road	Signal	1.78	A	9.1
Perry Highway & Graham Road	Signal	1.81	D	44.5
Sewickley Oakmont Road & Lombardy Drive	Signal	1.85	D	44.5

Intersection	Control	Dist (mi)	Exist LOS	Delay (s/veh)
Perrysville Avenue & Nelson Run Road	Signal	1.89	C	21.9
Babcock Boulevard & Browns Lane	Signal	1.90	B	11.3
Babcock Boulevard & Roderick Drive	Signal	1.95	D	42.1

5.0 TRIP GENERATION ANALYSIS

Trip generation for the proposed McKnight Road Retail Development was estimated using public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for Land Use Code 820 (Shopping Center / Retail). A 30% pass-by credit and 0% internal capture credit are applied per methodology meeting agreement, yielding the following net new external trips:

Land Use Code	820 — Shopping Center / Retail
Development size	200 1,000 sqft GLA
Weekday daily (gross)	16,000
AM peak hour (net new)	640 trips (390 in / 250 out)
PM peak hour (net new)	1,600 trips (768 in / 832 out)
Pass-by credit applied	30%
Internal capture applied	0%
Transit/mode reduction	Not applied

5.1 Trip Generation by Period

Period	Gross Trips	Pass-by	Int. Capture	Net External	In	Out
AM Peak Hour	640	48	0	455	277	178
PM Peak Hour	1,600	480	0	860	413	447
Saturday Midday	1,760	132	0	1,250	625	625
Daily Total	16,000	1,200	0	11,364	5,682	5,682

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross – pass-by – internal-capture) × auto-mode share (76.9% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

Trip Distribution by Time of Day

Estimated within-day distribution of the 16,000 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

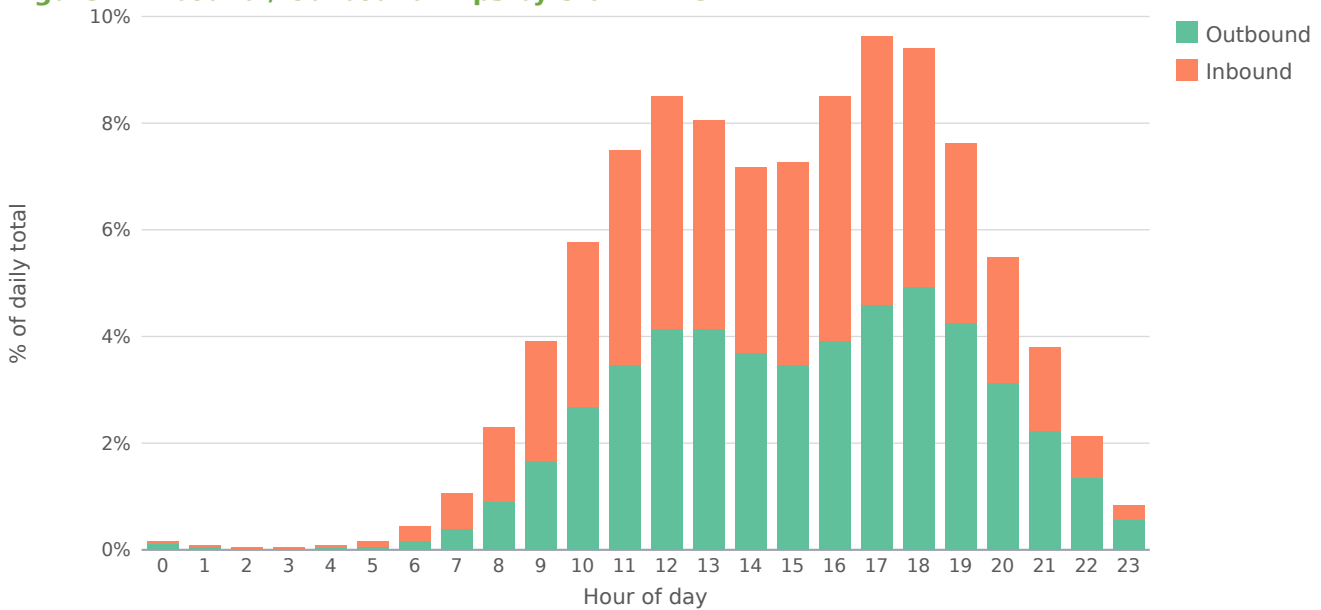
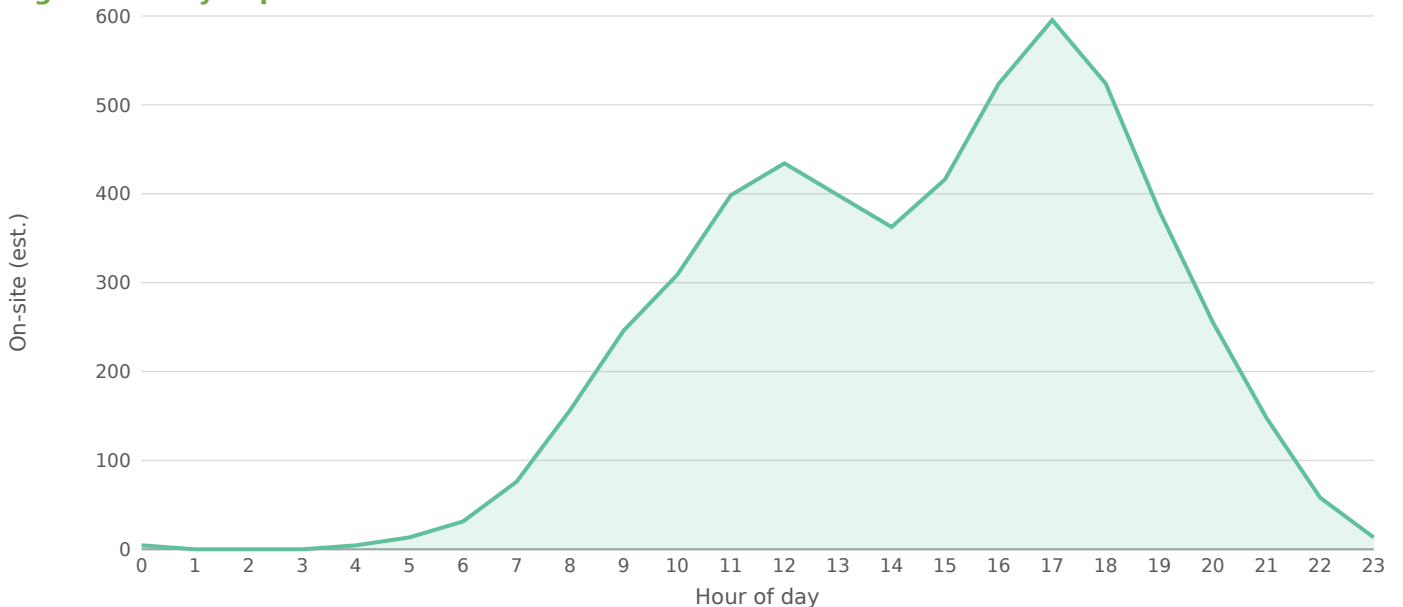


Figure — Daily Trip Accumulation



Peak on-site accumulation ~595 at 17:00, from 16,000 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

6.0 TRIP DISTRIBUTION AND ASSIGNMENT

Project-generated trips are distributed to the roadway network based on the existing directional distribution observed at the study-area count locations, proximity to the site access points, and the regional travel-demand model (PennDOT traffic count data).

Distribution method

Existing traffic directional distribution + engineering judgment

Assignment method

Site-access-proportional; confirmed at methodology meeting

Directional distribution percentages and trip assignment maps should be verified at the methodology meeting with the reviewing agency and attached as Appendix B.

6.1 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^1 \cdot d_{\text{site}})$); mass basis: intersection

through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	16%
Southeast (ESE + SSE)	21%
Southwest (SSW + WSW)	29%
Northwest (WNW + NNW)	33%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near McKnight Road	ENE	0.38	1,657	0.1	5.19%
McKnight Road & Braunlich Drive	ESE	0.44	1,657	0.1	5.08%
McKnight Road & Hilands Place	ESE	0.58	1,657	0.0	4.85%
Cemetery Lane & Babcock Boulevard	SSE	0.66	1,607	0.0	4.58%
Perrysville Avenue & West View Avenue	SSW	1.78	2,202	0.0	4.41%
McKnight Road & Ross Park Mall Drive	NNE	1.17	1,791	0.0	4.34%
McKnight Road & Patrick Place	NNE	0.87	1,624	0.0	4.33%
McKnight Road & Johnanna Drive	NNW	1.45	1,791	0.0	3.97%
McKnight Road & McKnight Circle	NNW	1.19	1,618	0.0	3.90%
McKnight Road & McIntyre Road	NNW	1.78	1,791	0.0	3.59%
Babcock Boulevard & Rochester Road	SSW	0.40	1,111	0.0	3.45%
Perrysville Avenue & Cemetery Lane	SSW	1.62	1,635	0.0	3.44%
Babcock Boulevard & Siebert Road	NNW	0.44	1,111	0.0	3.40%
Three Degree Road & Babcock Boulevard	NNW	1.68	1,618	0.0	3.35%
Babcock Boulevard & Thompson Run Road	ESE	1.39	1,397	0.0	3.16%
Babcock Boulevard & Browns Lane	NNW	1.90	1,618	0.0	3.13%
Babcock Boulevard & Roderick Drive	NNW	1.95	1,618	0.0	3.08%
Perry Highway & West View Park Drive	WSW	1.07	1,007	0.0	2.52%
Center Avenue & Perry Highway	WSW	1.08	1,007	0.0	2.51%
Perry Highway & Chalfonte Avenue	WSW	1.12	1,007	0.0	2.48%

36 study-area zone(s) (top 20 by trip share shown). Screening-grade distribution; not a calibrated regional model.

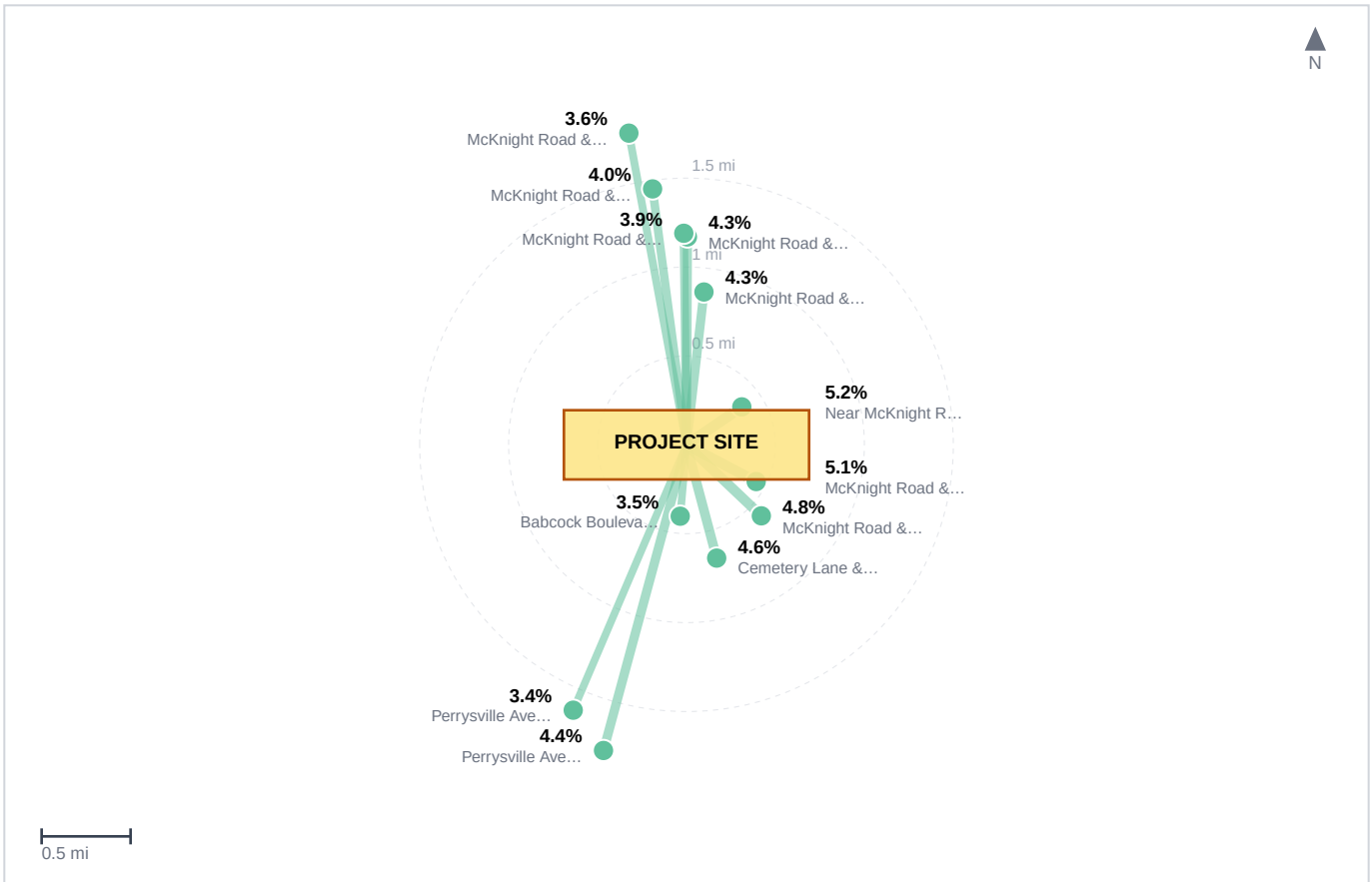
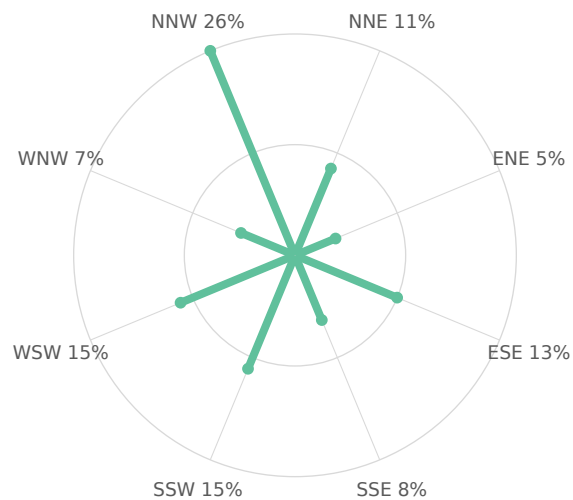


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

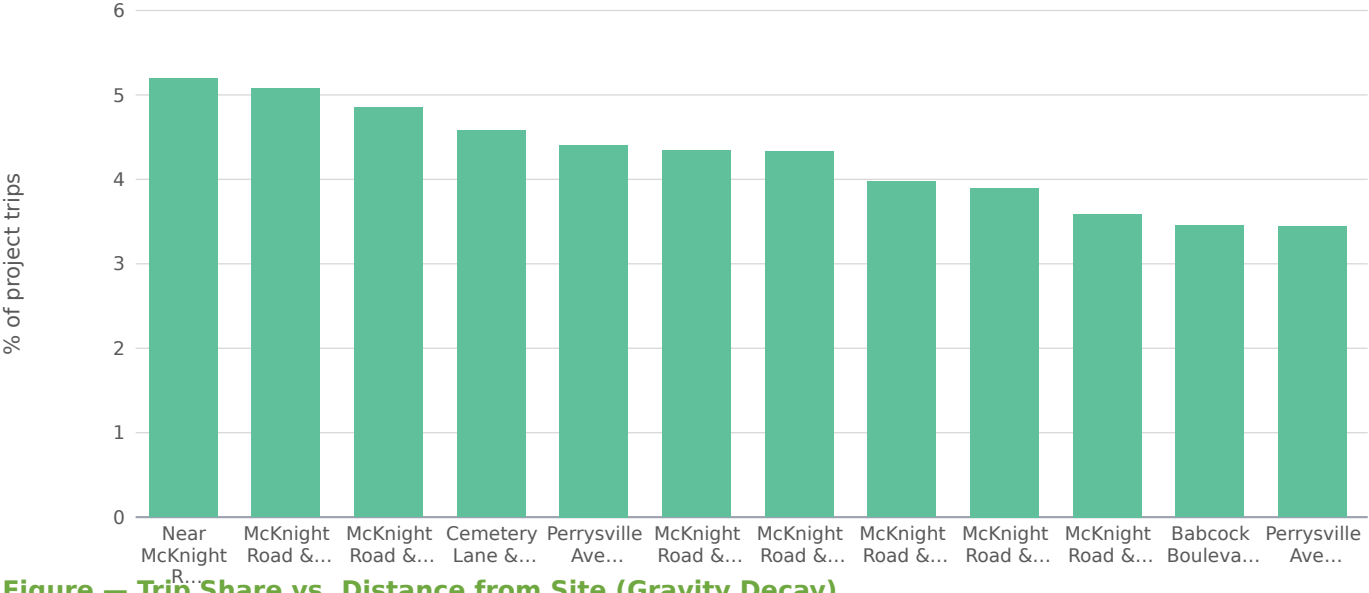
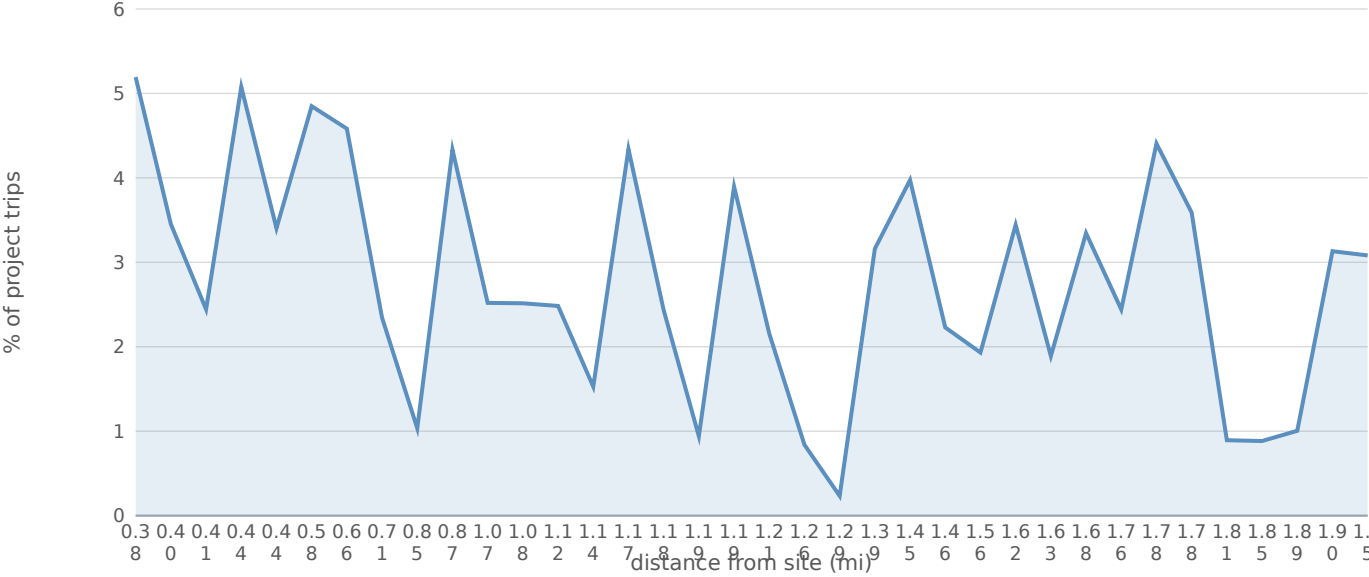


Figure — Trip Share vs. Distance from Site (Gravity Decay)



6.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near McKnight Road	ENE	0.38	69	130	4.8%
Babcock Boulevard & Rochester Road	SSW	0.40	66	112	4.2%
Siebert Road & McKnight Road	NNE	0.41	156	232	8.6%
McKnight Road & Braunlich Drive	ESE	0.44	50	94	3.5%
Babcock Boulevard & Siebert Road	NNW	0.44	99	205	7.6%
McKnight Road & Hilands Place	ESE	0.58	25	47	1.7%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Cemetery Lane & Babcock Boulevard	SSE	0.66	66	112	4.2%
Glenmore Avenue & Rochester Road	WSW	0.71	41	78	2.9%
Babcock Boulevard & Evergreen Road	SSE	0.85	6	12	0.4%
McKnight Road & Patrick Place	NNE	0.87	37	93	3.5%
Perry Highway & West View Park Drive	WSW	1.07	2	4	0.1%
Center Avenue & Perry Highway	WSW	1.08	2	4	0.1%
Perry Highway & Chalfonte Avenue	WSW	1.12	2	3	0.1%
Perry Highway & Highland Avenue	WSW	1.14	2	3	0.1%
McKnight Road & Ross Park Mall Drive	NNE	1.17	37	93	3.5%
Bellevue Road & Perry Highway	WSW	1.18	1	2	0.1%
Perry Highway & Rochester Road	WNW	1.19	41	78	2.9%
McKnight Road & McKnight Circle	NNW	1.19	101	151	5.6%
Perry Highway & Perry Lane	WNW	1.21	1	2	0.1%
Center Avenue & West View Park Drive	WSW	1.26	1	2	0.1%
McKnight Road & Nelson Run Road	SSE	1.29	56	140	5.2%
Babcock Boulevard & Thompson Run Road	ESE	1.39	33	49	1.8%
McKnight Road & Johnanna Drive	NNW	1.45	37	93	3.5%
Perrysville Avenue & Perrysville Road	SSW	1.46	0	1	0.0%
Perry Highway & Good Lane	WNW	1.56	0	0	0.0%
Perrysville Avenue & Cemetery Lane	SSW	1.62	33	63	2.3%
Perry Highway & Lindley Lane	WNW	1.63	0	0	0.0%
Three Degree Road & Babcock Boulevard	NNW	1.68	99	205	7.6%
Peoples Road & Babcock Boulevard	SSE	1.76	33	49	1.8%
Perrysville Avenue & West View Avenue	SSW	1.78	33	63	2.3%
McKnight Road & McIntyre Road	NNW	1.78	37	93	3.5%
Perry Highway & Graham Road	NNW	1.81	71	135	5.0%
Sewickley Oakmont Road & Lombardy Drive	NNW	1.85	71	135	5.0%
Perrysville Avenue & Nelson Run Road	SSW	1.89	33	63	2.3%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Babcock Boulevard & Browns Lane	NNW	1.90	28	70	2.6%
Babcock Boulevard & Roderick Drive	NNW	1.95	28	70	2.6%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

7.0 FUTURE CONDITIONS — NO-BUILD

The No-Build scenario represents future traffic volumes in the 2028 without the proposed development. Background traffic growth is applied at 1.50% per year (compound) over 2 years to current counts. Programmed improvements from the PennDOT Statewide Transportation Improvement Program (STIP) and local capital programs are incorporated into the No-Build network where construction is funded and committed within the analysis horizon.

No-Build horizon year	2028
Growth applied	1.50%/yr × 2 yr
Programmed projects	Per STIP and local CIP — confirm at methodology meeting

8.0 FUTURE CONDITIONS — BUILD

The Build scenario adds distributed project-generated trips to the No-Build network for the 2028 analysis horizon. All 36 study-area intersections and roadway segments are evaluated for LOS under Build conditions.

8.1 Intersection Level of Service — Three-Scenario Summary

Intersection	Ctrl	Dist (mi)	Trips	Exist LOS	No-Bld LOS	Build LOS	Δ delay (s)	Q95 (ft)
Near McKnight Road	Sig	0.38	130	C	C	D	14.2	27
Babcock Boulevard & Rochester Road	Sig	0.40	112	D	D	D	12.8	53
Siebert Road & McKnight Road	Sig	0.41	232	B	B	B	2.1	33
McKnight Road & Braunlich Drive	Sig	0.44	94	A	A	A	0.6	49
Babcock Boulevard & Siebert Road	Sig	0.44	205	D	D	E	28.0	59
McKnight Road & Hilands Place	Sig	0.58	47	C	C	C	1.9	28
Cemetery Lane & Babcock Boulevard	Sig	0.66	112	D	D	E	12.2	1,00
Glenmore Avenue & Rochester Road	Sig	0.71	78	C	C	C	4.7	24
Babcock Boulevard & Evergreen Road	Sig	0.85	12	B	B	B	0.1	5
McKnight Road & Patrick Place	Sig	0.87	93	A	A	A	0.5	55
Perry Highway & West View Park Drive	Sig	1.07	4	A	A	A	0.0	20
Center Avenue & Perry Highway	Sig	1.08	4	B	B	B	0.1	25
Perry Highway & Chalfonte Avenue	Sig	1.12	3	C	C	C	0.1	25
Perry Highway & Highland Avenue	Sig	1.14	3	F	F	F	0.9	23
McKnight Road & Ross Park Mall Drive	Sig	1.17	93	A	A	A	0.6	73
Bellevue Road & Perry Highway	Sig	1.18	2	D	D	D	0.2	30
Perry Highway & Rochester Road	Sig	1.19	78	B	B	B	0.5	10

Intersection	Ctrl	Dist (mi)	Trips	Exist LOS	No-Bld LOS	Build LOS	Δ delay (s)	Q95 (ft)
McKnight Road & McKnight Circle	Sig	1.19	151	A	A	B	0.9	62
Perry Highway & Perry Lane	Sig	1.21	2	C	C	C	0.1	22
Center Avenue & West View Park Drive	Sig	1.26	2	A	A	A	0.0	5
McKnight Road & Nelson Run Road	Sig	1.29	140	A	A	A	0.3	5
Babcock Boulevard & Thompson Run Road	Sig	1.39	49	D	D	D	4.5	75
McKnight Road & Johnanna Drive	Sig	1.45	93	B	B	B	1.0	77
Perrysville Avenue & Perrysville Road	Sig	1.46	1	B	B	B	0.0	13
Perry Highway & Good Lane	Sig	1.56	0	C	C	C	0.0	27
Perrysville Avenue & Cemetery Lane	Sig	1.62	63	F	F	F	15.0	89
Perry Highway & Lindley Lane	Sig	1.63	0	D	D	D	0.0	30
Three Degree Road & Babcock Boulevard	Sig	1.68	205	F	F	F	48.4	93
Peoples Road & Babcock Boulevard	Sig	1.76	49	F	F	F	13.9	93
Perrysville Avenue & West View Avenue	Sig	1.78	63	F	F	F	15.4	1,19
McKnight Road & McIntyre Road	Sig	1.78	93	A	A	A	0.6	77
Perry Highway & Graham Road	Sig	1.81	135	D	D	E	24.2	62
Sewickley Oakmont Road & Lombardy Drive	Sig	1.85	135	D	D	E	24.2	57
Perrysville Avenue & Nelson Run Road	Sig	1.89	63	C	C	C	0.9	85
Babcock Boulevard & Browns Lane	Sig	1.90	70	B	B	B	0.7	49
Babcock Boulevard & Roderick Drive	Sig	1.95	70	D	D	D	7.3	1,17

Acceptable PennDOT LOS threshold: LOS D (urban) / LOS C (rural). Locations shown at Build LOS E or F require mitigation analysis in §9.

Lane-Group Queues at Intersections with Measured Turning Movements

At the intersections below, an imported Synchro record supplied measured turning-movement counts, so the background traffic carries a real left / through / right split rather than an assumed one. Each lane group is analyzed on the same one-critical-lane screening basis as the approach (saturation flow × g/C), with the project's own trips assigned to specific movements by the path assignment; lane-group volumes cross-foot to their approach total. Where the imported record also carried a bay length, the queue is compared against that bay and a deficit is flagged. Where the record also carried LANE COUNTS, the Lanes column shows them and that group's capacity is sized as lanes × saturation flow × g/C; an em-dash in the Lanes column means no count was supplied and the group falls back to a single critical lane. Signal phasing is not carried by the import, so a calibrated Synchro / SimTraffic run supersedes these figures for design.

Intersection	Lane group	Lanes	Build vph	v/c	Q95 (ft)	Bay (ft)	Storage
Babcock Boulevard & Rochester Road	NB L	—	180	0.75	149	—	—
Babcock Boulevard & Rochester Road	NB T	—	471	0.64	324	—	—
Babcock Boulevard & Rochester Road	NB R	—	0	0.00	0	—	—
Babcock Boulevard & Rochester Road	SB L	—	0	0.00	0	—	—
Babcock Boulevard & Rochester Road	SB T	—	425	0.58	283	—	—
Babcock Boulevard & Rochester Road	SB R	—	180	0.24	102	—	—
Babcock Boulevard & Rochester Road	EB L	—	180	0.39	128	—	—
Babcock Boulevard & Rochester Road	EB T	—	0	0.00	0	—	—
Babcock Boulevard & Rochester Road	EB R	—	180	0.39	128	—	—
Siebert Road & McKnight Road	NB L	—	0	0.00	0	—	—

Siebert Road & McKnight Road	NB T	—	0	0.00	0	— —
Siebert Road & McKnight Road	NB R	—	0	0.00	0	— —
Siebert Road & McKnight Road	SB L	—	0	0.00	0	— —
Siebert Road & McKnight Road	SB T	—	349	0.44	166	— —
Siebert Road & McKnight Road	SB R	—	209	0.26	91	— —
Siebert Road & McKnight Road	EB L	—	0	0.00	0	— —
Siebert Road & McKnight Road	EB T	—	0	0.00	0	— —
Siebert Road & McKnight Road	EB R	—	0	0.00	0	— —
Siebert Road & McKnight Road	WB L	—	57	0.08	25	— —
Siebert Road & McKnight Road	WB T	—	384	0.54	204	— —
Siebert Road & McKnight Road	WB R	—	0	0.00	0	— —
McKnight Road & Braunlich Drive	NB L	—	14	0.01	4	— —
McKnight Road & Braunlich Drive	NB T	—	0	0.00	0	— —
McKnight Road & Braunlich Drive	NB R	—	0	0.00	0	— —
McKnight Road & Braunlich Drive	SB L	—	360	0.34	136	— —
McKnight Road & Braunlich Drive	SB T	—	493	0.46	205	— —
McKnight Road & Braunlich Drive	SB R	—	0	0.00	0	— —
McKnight Road & Braunlich Drive	WB L	—	361	0.80	251	— —
McKnight Road & Braunlich Drive	WB T	—	31	0.07	18	— —
McKnight Road & Braunlich Drive	WB R	—	0	0.00	0	— —
Babcock Boulevard & Siebert Road	NB L	—	0	0.00	0	— —
Babcock Boulevard & Siebert Road	NB T	—	532	0.72	384	— —
Babcock Boulevard & Siebert Road	NB R	—	180	0.24	102	— —
Babcock Boulevard & Siebert Road	SB L	—	180	0.75	149	— —
Babcock Boulevard & Siebert Road	SB T	—	457	0.62	311	— —
Babcock Boulevard & Siebert Road	SB R	—	0	0.00	0	— —
Babcock Boulevard & Siebert Road	WB L	—	180	0.39	128	— —
Babcock Boulevard & Siebert Road	WB T	—	0	0.00	0	— —
Babcock Boulevard & Siebert Road	WB R	—	180	0.39	128	— —
Boulevard Cemetery Lane & Babcock	NB L	—	210	0.51	242	— —
Boulevard Cemetery Lane & Babcock	NB T	—	0	0.00	0	— —
Boulevard Cemetery Lane & Babcock	NB R	—	180	0.44	204	— —
Boulevard Cemetery Lane & Babcock	EB L	—	0	0.00	0	— —
Boulevard Cemetery Lane & Babcock	EB T	—	648	0.68	632	— —
Boulevard Cemetery Lane & Babcock	EB R	—	213	0.23	151	— —
Boulevard Cemetery Lane & Babcock	WB L	—	180	0.87	234	— —
Boulevard Cemetery Lane & Babcock	WB T	—	697	0.74	710	— —
Boulevard Cemetery Lane & Babcock	WB R	—	0	0.00	0	— —
Glenmore Avenue & Rochester Road	NB L	—	59	0.07	23	— —
Glenmore Avenue & Rochester Road	NB T	—	312	0.38	142	— —
Glenmore Avenue & Rochester Road	NB R	—	59	0.07	23	— —
Glenmore Avenue & Rochester Road	SB L	—	59	0.07	23	— —
Glenmore Avenue & Rochester Road	SB T	—	312	0.38	142	— —
Glenmore Avenue & Rochester Road	SB R	—	100	0.12	40	— —
Glenmore Avenue & Rochester Road	EB L	—	96	0.14	43	— —
Glenmore Avenue & Rochester Road	EB T	—	243	0.35	120	— —
Glenmore Avenue & Rochester Road	EB R	—	59	0.09	26	— —

Glenmore Avenue & Rochester Road	WB L	—	59	0.09	26	— —
Glenmore Avenue & Rochester Road	WB T	—	243	0.35	120	— —
Glenmore Avenue & Rochester Road	WB R	—	59	0.09	26	— —
McKnight Road & Patrick Place	NB L	—	0	0.00	0	— —
McKnight Road & Patrick Place	NB T	—	848	0.81	461	— —
McKnight Road & Patrick Place	NB R	—	81	0.08	24	— —
McKnight Road & Patrick Place	SB L	—	0	0.00	0	— —
McKnight Road & Patrick Place	SB T	—	0	0.00	0	— —
McKnight Road & Patrick Place	SB R	—	0	0.00	0	— —
McKnight Road & Patrick Place	EB L	—	81	0.18	44	— —
McKnight Road & Patrick Place	EB T	—	280	0.62	171	— —
McKnight Road & Patrick Place	EB R	—	0	0.00	0	— —
McKnight Road & Patrick Place	WB L	—	0	0.00	0	— —
McKnight Road & Patrick Place	WB T	—	0	0.00	0	— —
McKnight Road & Patrick Place	WB R	—	0	0.00	0	— —
Perry Highway & West View Park Drive	EB L	—	0	0.00	0	— —
Perry Highway & West View Park Drive	EB T	—	520	0.48	200	— —
Perry Highway & West View Park Drive	EB R	—	0	0.00	0	— —
Perry Highway & West View Park Drive	WB L	—	1	0.00	0	— —
Perry Highway & West View Park Drive	WB T	—	520	0.48	200	— —
Perry Highway & West View Park Drive	WB R	—	0	0.00	0	— —
Center Avenue & Perry Highway	NB L	—	64	0.07	23	— —
Center Avenue & Perry Highway	NB T	—	455	0.51	213	— —
Center Avenue & Perry Highway	NB R	—	1	0.00	0	— —
Center Avenue & Perry Highway	SB L	—	0	0.00	0	— —
Center Avenue & Perry Highway	SB T	—	455	0.51	213	— —
Center Avenue & Perry Highway	SB R	—	64	0.07	23	— —
Center Avenue & Perry Highway	EB L	—	0	0.00	0	— —
Center Avenue & Perry Highway	EB T	—	1	0.00	1	— —
Center Avenue & Perry Highway	EB R	—	0	0.00	0	— —
Center Avenue & Perry Highway	WB L	—	64	0.10	30	— —
Center Avenue & Perry Highway	WB T	—	235	0.38	122	— —
Center Avenue & Perry Highway	WB R	—	63	0.10	30	— —
Perry Highway & Chalfonte Avenue	NB L	—	65	0.07	23	— —
Perry Highway & Chalfonte Avenue	NB T	—	390	0.44	174	— —
Perry Highway & Chalfonte Avenue	NB R	—	65	0.07	23	— —
Perry Highway & Chalfonte Avenue	SB L	—	65	0.07	23	— —
Perry Highway & Chalfonte Avenue	SB T	—	390	0.44	174	— —
Perry Highway & Chalfonte Avenue	SB R	—	65	0.07	23	— —
Perry Highway & Chalfonte Avenue	EB L	—	64	0.10	30	— —
Perry Highway & Chalfonte Avenue	EB T	—	233	0.38	121	— —
Perry Highway & Chalfonte Avenue	EB R	—	64	0.10	30	— —
Perry Highway & Chalfonte Avenue	WB L	—	65	0.11	31	— —
Perry Highway & Chalfonte Avenue	WB T	—	233	0.38	121	— —
Perry Highway & Chalfonte Avenue	WB R	—	64	0.10	30	— —
Perry Highway & Highland Avenue	NB L	—	93	0.18	48	— —
Perry Highway & Highland Avenue	NB T	—	229	0.44	128	— —

Perry Highway & Highland Avenue	NB R	—	0	0.00	0	— —
Perry Highway & Highland Avenue	SB L	—	0	0.00	0	— —
Perry Highway & Highland Avenue	SB T	—	229	0.44	128	— —
Perry Highway & Highland Avenue	SB R	—	93	0.18	48	— —
Perry Highway & Highland Avenue	EB L	—	180	0.60	115	— —
Perry Highway & Highland Avenue	EB T	—	1	0.00	1	— —
Perry Highway & Highland Avenue	EB R	—	180	0.34	97	— —
Perry Highway & Highland Avenue	WB L	—	29	0.10	17	— —
Perry Highway & Highland Avenue	WB T	—	305	0.57	178	— —
Perry Highway & Highland Avenue	WB R	—	28	0.05	14	— —
McKnight Road & Ross Park Mall Drive	NB L	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	NB T	—	931	0.84	593	— —
McKnight Road & Ross Park Mall Drive	NB R	—	85	0.08	28	— —
McKnight Road & Ross Park Mall Drive	SB L	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	SB T	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	SB R	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	EB L	—	85	0.20	54	— —
McKnight Road & Ross Park Mall Drive	EB T	—	276	0.64	198	— —
McKnight Road & Ross Park Mall Drive	EB R	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	WB L	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	WB T	—	0	0.00	0	— —
McKnight Road & Ross Park Mall Drive	WB R	—	0	0.00	0	— —
Bellevue Road & Perry Highway	NB L	—	180	0.70	138	— —
Bellevue Road & Perry Highway	NB T	—	338	0.50	208	— —
Bellevue Road & Perry Highway	NB R	—	0	0.00	0	— —
Bellevue Road & Perry Highway	SB L	—	0	0.00	0	— —
Bellevue Road & Perry Highway	SB T	—	338	0.50	208	— —
Bellevue Road & Perry Highway	SB R	—	180	0.26	100	— —
Bellevue Road & Perry Highway	EB L	—	180	0.38	118	— —
Bellevue Road & Perry Highway	EB T	—	1	0.00	1	— —
Bellevue Road & Perry Highway	EB R	—	180	0.38	118	— —
Perry Highway & Rochester Road	NB L	—	40	0.07	20	— —
Perry Highway & Rochester Road	NB T	—	122	0.23	63	— —
Perry Highway & Rochester Road	NB R	—	40	0.07	20	— —
Perry Highway & Rochester Road	SB L	—	40	0.07	20	— —
Perry Highway & Rochester Road	SB T	—	122	0.23	63	— —
Perry Highway & Rochester Road	SB R	—	40	0.07	20	— —
Perry Highway & Rochester Road	EB L	—	40	0.04	13	— —
Perry Highway & Rochester Road	EB T	—	318	0.33	124	— —
Perry Highway & Rochester Road	EB R	—	40	0.04	13	— —
Perry Highway & Rochester Road	WB L	—	40	0.04	13	— —
Perry Highway & Rochester Road	WB T	—	322	0.33	125	— —
Perry Highway & Rochester Road	WB R	—	40	0.04	13	— —
McKnight Road & McKnight Circle	NB L	—	0	0.00	0	— —
McKnight Road & McKnight Circle	NB T	—	0	0.00	0	— —
McKnight Road & McKnight Circle	NB R	—	0	0.00	0	— —
McKnight Road & McKnight Circle	SB L	—	0	0.00	0	— —

McKnight Road & McKnight Circle	SB T	—	903	0.86	522	— —
McKnight Road & McKnight Circle	SB R	—	81	0.08	24	— —
McKnight Road & McKnight Circle	EB L	—	0	0.00	0	— —
McKnight Road & McKnight Circle	EB T	—	0	0.00	0	— —
McKnight Road & McKnight Circle	EB R	—	0	0.00	0	— —
McKnight Road & McKnight Circle	WB L	—	81	0.18	44	— —
McKnight Road & McKnight Circle	WB T	—	280	0.62	170	— —
McKnight Road & McKnight Circle	WB R	—	0	0.00	0	— —
Perry Highway & Perry Lane	NB L	—	61	0.07	23	— —
Perry Highway & Perry Lane	NB T	—	341	0.40	154	— —
Perry Highway & Perry Lane	NB R	—	61	0.07	23	— —
Perry Highway & Perry Lane	SB L	—	61	0.07	23	— —
Perry Highway & Perry Lane	SB T	—	341	0.40	154	— —
Perry Highway & Perry Lane	SB R	—	61	0.07	23	— —
Perry Highway & Perry Lane	EB L	—	61	0.09	28	— —
Perry Highway & Perry Lane	EB T	—	240	0.36	121	— —
Perry Highway & Perry Lane	EB R	—	61	0.09	28	— —
Perry Highway & Perry Lane	WB L	—	61	0.09	28	— —
Perry Highway & Perry Lane	WB T	—	240	0.36	121	— —
Perry Highway & Perry Lane	WB R	—	61	0.09	28	— —
Center Avenue & West View Park Drive	NB L	—	0	0.00	0	— —
Center Avenue & West View Park Drive	NB T	—	0	0.00	0	— —
Center Avenue & West View Park Drive	NB R	—	0	0.00	0	— —
Center Avenue & West View Park Drive	EB L	—	0	0.00	0	— —
Center Avenue & West View Park Drive	EB T	—	1	0.00	0	— —
Center Avenue & West View Park Drive	EB R	—	0	0.00	0	— —
Center Avenue & West View Park Drive	WB L	—	122	0.11	36	— —
Center Avenue & West View Park Drive	WB T	—	63	0.06	18	— —
Center Avenue & West View Park Drive	WB R	—	0	0.00	0	— —
McKnight Road & Nelson Run Road	NB L	—	0	0.00	0	— —
McKnight Road & Nelson Run Road	NB T	—	0	0.00	0	— —
McKnight Road & Nelson Run Road	NB R	—	0	0.00	0	— —
McKnight Road & Nelson Run Road	WB L	—	192	0.18	59	— —
McKnight Road & Nelson Run Road	WB T	—	0	0.00	0	— —
McKnight Road & Nelson Run Road	WB R	—	0	0.00	0	— —
Babcock Boulevard & Thompson Run Road	SB L	—	180	0.42	166	— —
Babcock Boulevard & Thompson Run Road	SB T	—	0	0.00	0	— —
Babcock Boulevard & Thompson Run Road	SB R	—	180	0.42	166	— —
Babcock Boulevard & Thompson Run Road	EB L	—	180	0.83	192	— —
Babcock Boulevard & Thompson Run Road	EB T	—	539	0.62	435	— —
Babcock Boulevard & Thompson Run Road	EB R	—	0	0.00	0	— —
Babcock Boulevard & Thompson Run Road	WB L	—	0	0.00	0	— —
Babcock Boulevard & Thompson Run Road	WB T	—	588	0.68	494	— —
Babcock Boulevard & Thompson Run Road	WB R	—	180	0.21	113	— —
McKnight Road & Johnanna Drive	NB L	—	88	0.08	28	— —
McKnight Road & Johnanna Drive	NB T	—	840	0.76	485	— —
McKnight Road & Johnanna Drive	NB R	—	88	0.08	28	— —

McKnight Road & Johnanna Drive	SB L	—	0	0.00	0	— —
McKnight Road & Johnanna Drive	SB T	—	0	0.00	0	— —
McKnight Road & Johnanna Drive	SB R	—	0	0.00	0	— —
McKnight Road & Johnanna Drive	EB L	—	88	0.20	56	— —
McKnight Road & Johnanna Drive	EB T	—	273	0.63	196	— —
McKnight Road & Johnanna Drive	EB R	—	0	0.00	0	— —
McKnight Road & Johnanna Drive	WB L	—	0	0.00	0	— —
McKnight Road & Johnanna Drive	WB T	—	273	0.63	196	— —
McKnight Road & Johnanna Drive	WB R	—	88	0.20	56	— —
Perry Highway & Good Lane	NB L	—	0	0.00	0	— —
Perry Highway & Good Lane	NB T	—	333	0.53	182	— —
Perry Highway & Good Lane	NB R	—	130	0.20	62	— —
Perry Highway & Good Lane	SB L	—	130	0.43	80	— —
Perry Highway & Good Lane	SB T	—	333	0.53	182	— —
Perry Highway & Good Lane	SB R	—	0	0.00	0	— —
Perry Highway & Good Lane	WB L	—	0	0.00	0	— —
Perry Highway & Good Lane	WB T	—	0	0.00	0	— —
Perry Highway & Good Lane	WB R	—	0	0.00	0	— —
Perrysville Avenue & Cemetery Lane	NB L	—	0	0.00	0	— —
Perrysville Avenue & Cemetery Lane	NB T	—	784	0.97	789	— —
Perrysville Avenue & Cemetery Lane	NB R	—	88	0.11	52	— —
Perrysville Avenue & Cemetery Lane	SB L	—	58	0.07	34	— —
Perrysville Avenue & Cemetery Lane	SB T	—	784	0.97	789	— —
Perrysville Avenue & Cemetery Lane	SB R	—	0	0.00	0	— —
Perrysville Avenue & Cemetery Lane	EB L	—	196	0.24	125	— —
Perrysville Avenue & Cemetery Lane	EB T	—	381	0.47	274	— —
Perrysville Avenue & Cemetery Lane	EB R	—	196	0.24	125	— —
Perrysville Avenue & Cemetery Lane	WB L	—	213	0.26	137	— —
Perrysville Avenue & Cemetery Lane	WB T	—	0	0.00	0	— —
Perrysville Avenue & Cemetery Lane	WB R	—	180	0.22	114	— —
Perry Highway & Lindley Lane	NB L	—	180	0.65	126	— —
Perry Highway & Lindley Lane	NB T	—	283	0.45	164	— —
Perry Highway & Lindley Lane	NB R	—	0	0.00	0	— —
Perry Highway & Lindley Lane	SB L	—	0	0.00	0	— —
Perry Highway & Lindley Lane	SB T	—	283	0.45	164	— —
Perry Highway & Lindley Lane	SB R	—	180	0.29	98	— —
Perry Highway & Lindley Lane	EB L	—	180	0.37	109	— —
Perry Highway & Lindley Lane	EB T	—	0	0.00	0	— —
Perry Highway & Lindley Lane	EB R	—	180	0.37	109	— —
Three Degree Road & Babcock Boulevard	NB L	—	328	0.40	227	— —
Three Degree Road & Babcock Boulevard	NB T	—	0	0.00	0	— —
Three Degree Road & Babcock Boulevard	NB R	—	328	0.40	227	— —
Three Degree Road & Babcock Boulevard	EB L	—	0	0.00	0	— —
Three Degree Road & Babcock Boulevard	EB T	—	576	0.71	480	— —
Three Degree Road & Babcock Boulevard	EB R	—	323	0.40	223	— —
Three Degree Road & Babcock Boulevard	WB L	—	258	0.32	171	— —
Three Degree Road & Babcock Boulevard	WB T	—	576	0.71	480	— —

Three Degree Road & Babcock Boulevard	WB R	—	0	0.00	0	— —
Peoples Road & Babcock Boulevard	NB L	—	365	0.56	401	— —
Peoples Road & Babcock Boulevard	NB T	—	0	0.00	0	— —
Peoples Road & Babcock Boulevard	NB R	—	258	0.39	263	— —
Peoples Road & Babcock Boulevard	EB L	—	0	0.00	0	— —
Peoples Road & Babcock Boulevard	EB T	—	307	0.47	324	— —
Peoples Road & Babcock Boulevard	EB R	—	365	0.56	401	— —
Peoples Road & Babcock Boulevard	WB L	—	257	0.95	351	— —
Peoples Road & Babcock Boulevard	WB T	—	258	0.39	264	— —
Peoples Road & Babcock Boulevard	WB R	—	0	0.00	0	— —
Perryville Avenue & West View Avenue	SB L	—	180	0.22	114	— —
Perryville Avenue & West View Avenue	SB T	—	0	0.00	0	— —
Perryville Avenue & West View Avenue	SB R	—	180	0.22	114	— —
Perryville Avenue & West View Avenue	EB L	—	180	0.22	114	— —
Perryville Avenue & West View Avenue	EB T	—	987	1.22	1,010	— —
Perryville Avenue & West View Avenue	EB R	—	0	0.00	0	— —
Perryville Avenue & West View Avenue	WB L	—	0	0.00	0	— —
Perryville Avenue & West View Avenue	WB T	—	984	1.21	1,007	— —
Perryville Avenue & West View Avenue	WB R	—	180	0.22	114	— —
McKnight Road & McIntyre Road	NB L	—	0	0.00	0	— —
McKnight Road & McIntyre Road	NB T	—	931	0.84	593	— —
McKnight Road & McIntyre Road	NB R	—	85	0.08	28	— —
McKnight Road & McIntyre Road	SB L	—	0	0.00	0	— —
McKnight Road & McIntyre Road	SB T	—	0	0.00	0	— —
McKnight Road & McIntyre Road	SB R	—	0	0.00	0	— —
McKnight Road & McIntyre Road	EB L	—	85	0.20	54	— —
McKnight Road & McIntyre Road	EB T	—	276	0.64	198	— —
McKnight Road & McIntyre Road	EB R	—	0	0.00	0	— —
McKnight Road & McIntyre Road	WB L	—	0	0.00	0	— —
McKnight Road & McIntyre Road	WB T	—	0	0.00	0	— —
McKnight Road & McIntyre Road	WB R	—	0	0.00	0	— —
Perry Highway & Graham Road	NB L	—	2	0.00	1	— —
Perry Highway & Graham Road	NB T	—	22	0.06	13	— —
Perry Highway & Graham Road	NB R	—	208	0.54	138	— —
Perry Highway & Graham Road	SB L	—	273	0.71	188	— —
Perry Highway & Graham Road	SB T	—	22	0.06	13	— —
Perry Highway & Graham Road	SB R	—	2	0.00	1	— —
Perry Highway & Graham Road	EB L	—	2	0.01	1	— —
Perry Highway & Graham Road	EB T	—	357	0.49	198	— —
Perry Highway & Graham Road	EB R	—	2	0.00	1	— —
Perry Highway & Graham Road	WB L	—	208	0.75	148	— —
Perry Highway & Graham Road	WB T	—	357	0.49	199	— —
Perry Highway & Graham Road	WB R	—	278	0.38	146	— —
Sewickley Oakmont Road & Lombardy Drive	NB L	—	208	0.54	138	— —
Sewickley Oakmont Road & Lombardy Drive	NB T	—	92	0.24	57	— —
Sewickley Oakmont Road & Lombardy Drive	NB R	—	2	0.00	1	— —
Sewickley Oakmont Road & Lombardy Drive	SB L	—	2	0.00	1	— —

Sewickley Oakmont Road & Lombardy Drive	SB T	—	87	0.23	54	— —
Sewickley Oakmont Road & Lombardy Drive	SB R	—	208	0.54	138	— —
Sewickley Oakmont Road & Lombardy Drive	EB L	—	208	0.75	148	— —
Sewickley Oakmont Road & Lombardy Drive	EB T	—	357	0.49	199	— —
Sewickley Oakmont Road & Lombardy Drive	EB R	—	208	0.29	105	— —
Sewickley Oakmont Road & Lombardy Drive	WB L	—	2	0.01	1	— —
Sewickley Oakmont Road & Lombardy Drive	WB T	—	357	0.49	198	— —
Sewickley Oakmont Road & Lombardy Drive	WB R	—	2	0.00	1	— —
Perryville Avenue & Nelson Run Road	NB L	—	0	0.00	0	— —
Perryville Avenue & Nelson Run Road	NB T	—	0	0.00	0	— —
Perryville Avenue & Nelson Run Road	NB R	—	266	0.85	310	— —
Perryville Avenue & Nelson Run Road	SB L	—	299	0.95	356	— —
Perryville Avenue & Nelson Run Road	SB T	—	0	0.00	0	— —
Perryville Avenue & Nelson Run Road	SB R	—	0	0.00	0	— —
Perryville Avenue & Nelson Run Road	WB L	—	386	0.42	292	— —
Perryville Avenue & Nelson Run Road	WB T	—	0	0.00	0	— —
Perryville Avenue & Nelson Run Road	WB R	—	416	0.46	321	— —
Babcock Boulevard & Browns Lane	EB L	—	0	0.00	0	— —
Babcock Boulevard & Browns Lane	EB T	—	904	0.83	496	— —
Babcock Boulevard & Browns Lane	EB R	—	0	0.00	0	— —
Babcock Boulevard & Browns Lane	WB L	—	0	0.00	0	— —
Babcock Boulevard & Browns Lane	WB T	—	834	0.77	424	— —
Babcock Boulevard & Browns Lane	WB R	—	0	0.00	0	— —
Babcock Boulevard & Roderick Drive	NB L	—	180	0.87	244	— —
Babcock Boulevard & Roderick Drive	NB T	—	723	0.76	780	— —
Babcock Boulevard & Roderick Drive	NB R	—	0	0.00	0	— —
Babcock Boulevard & Roderick Drive	SB L	—	0	0.00	0	— —
Babcock Boulevard & Roderick Drive	SB T	—	653	0.68	662	— —
Babcock Boulevard & Roderick Drive	SB R	—	180	0.19	129	— —
Babcock Boulevard & Roderick Drive	EB L	—	180	0.44	212	— —
Babcock Boulevard & Roderick Drive	EB T	—	0	0.00	0	— —
Babcock Boulevard & Roderick Drive	EB R	—	180	0.44	212	— —

No lane group exceeds its imported bay storage under Build conditions.

9.0 MITIGATION ANALYSIS

The following mitigation measures are proposed to address Build-scenario LOS deficiencies identified in §8. Mitigation must be incorporated as permit conditions in the PennDOT access permit and local site-plan approval.

9.1 Signal Timing Optimization

Signal retiming at affected intersections is proposed to improve cycle lengths and phase splits to accommodate the added project volume. Synchro timing plans should be submitted to the applicable signal operations authority (PennDOT or local traffic engineering) for review and implementation.

9.2 Geometric Improvements

Where signal retiming alone is insufficient to achieve acceptable LOS, geometric improvements (auxiliary turn lanes, channelization) may be required. Turn-lane warrants should be confirmed per the applicable design manual.

9.3 Pedestrian and Multimodal Improvements

Pedestrian crossing enhancements (high-emphasis crosswalks, ADA-compliant ramps per PROWAG)

at signalized intersections along the project frontage roads should be coordinated with the applicable agency traffic engineering division.

9.4 Storage Bay Adequacy

95th-percentile queue analysis was performed at driveways and adjacent signalized intersections. Storage bay adequacy was evaluated against projected Build-scenario left-turn and right-turn queue lengths. Where storage deficiencies exist (95th-percentile queue exceeds available storage), turn-lane extensions or new auxiliary lanes are identified as required mitigation.

No intersection carries a field-measured `existingStorageFt` value. Supply that field on each affected intersection record to produce a Required-vs-Existing-vs-Deficit table per AASHTO Green Book Chapter 9 / applicable state DOT design manual storage-bay standards.

9.5 State-Specific Requirements

- Highway Occupancy Permit (HOP) — PennDOT Form M-945A — required for any access to a state highway.
- PennDOT Pub 282 specifies mandatory report format, contents checklist, and LOS analysis procedures — follow the Pub 282 outline for PennDOT submittals.
- Sight-distance analysis per PennDOT Pub 70 (Engineering and Traffic Studies) or AASHTO Green Book required at all driveways.

10.0 SITE ACCESS AND INTERNAL CIRCULATION

Driveway access geometry, sight distance, and on-site circulation are reviewed per PennDOT Publication 282 — Guidelines for the Development and Review of Traffic Impact Studies (current edition); PennDOT Publication 46 — Traffic Engineering Manual and the applicable access management standards. Access permit requirements: PennDOT Pub 282 defines three categories: Category 1 (≥ 100 PHT) requires full TIS; Category 2 (50-99 PHT) requires abbreviated TIS; Category 3 (< 50 PHT) exempt from TIS but may require an operational analysis. Highway Occupancy Permit (HOP) required for any access to a state highway. The analysis horizon for access-permit purposes is Existing + opening year + 10-year design horizon per PennDOT Pub 282. A secondary 20-year horizon may be required for principal arterials..

Study horizons	Existing + opening year + 10-year design horizon per PennDOT Pub 282. A secondary 20-year horizon may be required for principal arterials.
Access permit authority	PennDOT
Sight distance standard	AASHTO Green Book / state design manual, per design speed

Site access geometry, driveways, internal circulation, and parking design are dependent on the final site plan. A detailed site-access analysis per the applicable design manual should be performed once the site plan is finalized.

11.0 TRANSIT AND MULTIMODAL CONSIDERATIONS

Existing transit service, bicycle facilities, and pedestrian infrastructure within 2.00 miles of the site were inventoried. Any transit-mode reduction applied to trip generation reflects the transit availability factor approved at the methodology meeting. Multimodal improvements (transit shelter upgrades, bicycle parking, pedestrian connections) should be coordinated with the applicable transit provider and included in the site-plan submittal.

Transit reduction applied	None
Nearest transit stop	Verify field
Bicycle facility	Verify field inventory

12.0 CONCLUSIONS AND RECOMMENDATIONS

The proposed McKnight Road Retail Development is projected to generate 1,600 net new external PM peak-hour trips. 6 intersections drop one or more LOS grades and 9 locations operate at LOS E or F under Build conditions. The mitigation measures identified in §9 are required as conditions of the PennDOT access permit to maintain acceptable operations.

Professional Engineer Certification

This Traffic Impact Study (TIS) has been prepared and reviewed by a Professional Engineer licensed in the State of Pennsylvania under Pennsylvania Engineering Practice Act (63 Pa. C.S. § 21 et seq. (Engineering Practice Act); 49 Pa. Code Ch. 37). The signing PE attests to the accuracy and completeness of this study to the degree specified by the controlling PennDOT guidelines.

PE Seal — Pennsylvania
Pennsylvania Engineering Practice Act
63 Pa. C.S. § 21 et seq. (Engineering Practice Act);
49 Pa. Code Ch. 37

Signature: _____

Date: _____

Appendices (referenced — not auto-generated)

- Appendix A — Methodology Meeting Notes / Scope Letter
- Appendix B — Trip Generation Worksheets (public-data rates)
- Appendix C — Traffic Count Data
- Appendix D — Intersection Capacity Analysis Output (HCS / Synchro)
- Appendix E — Signal Warrant Analyses (where applicable)
- Appendix F — Crash Data (3-year, where required by reviewing agency)

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	200 1,000 sqft GLA
Daily trips (productions)	16,000
PM peak-hour trips	1,600 (768 in / 832 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{-(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Siebert Road & McKnight Road	0.41	1.0	232	8.6%
Babcock Boulevard & Siebert Road	0.44	1.1	205	7.6%
Three Degree Road & Babcock Boulevard	1.68	4.0	205	7.6%
McKnight Road & McKnight Circle	1.19	2.9	151	5.6%
McKnight Road & Nelson Run Road	1.29	3.1	140	5.2%
Perry Highway & Graham Road	1.81	4.3	135	5.0%
Sewickley Oakmont Road & Lombardy Drive	1.85	4.4	135	5.0%
Near McKnight Road	0.38	0.9	130	4.8%
Babcock Boulevard & Rochester Road	0.40	1.0	112	4.2%
Cemetery Lane & Babcock Boulevard	0.66	1.6	112	4.2%
McKnight Road & Braunlich Drive	0.44	1.1	94	3.5%
McKnight Road & Patrick Place	0.87	2.1	93	3.5%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 77% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 23% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.60. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	5.52	860	0.60
Minor arterial	3.99	312	0.31
Trunk highway	3.43	173	0.20
Principal arterial	3.50	53	0.16

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 10 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 1,134 network junctions, max imbalance 0.00 veh. 22 study intersection(s) carry path-derived movements; 14 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c , control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 36 signalized intersections lie within the 2-mile study area, and all 36 are carried as study intersections. The regional signal inventory holds 72 records inside that radius; 36 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

Background approach volumes are resolved per leg — the signal's design hour on the main road (its counted volume where the analyzer had a compatible count, else the road-class baseline it assigned — each worksheet says which), the road-class baseline on uncounted legs, client link counts where supplied; half per direction, and half in the physical direction of a one-way carriageway — and the background turning movements in the diagrams are balanced to the exit legs by iterative proportional fitting (NCHRP 255/765 refinement) from a geometry seed; each worksheet states its leg sources and residual. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

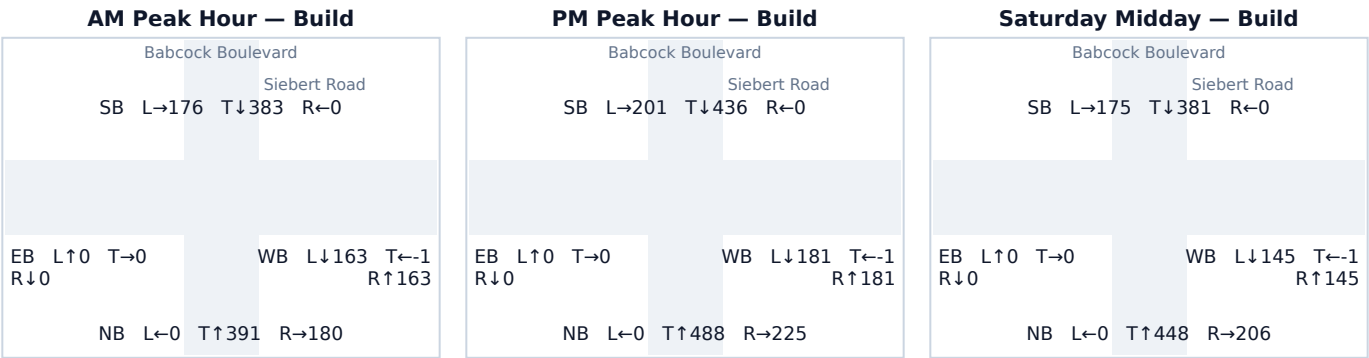
A.1 Babcock Boulevard & Siebert Road

Signal ID	pittsburgh-295
Location	NW Pittsburgh · 0.44 mi from site
Added PM peak trips	205
Existing / No-Build (PM)	LOS D · 39.6 s/veh · v/c 0.92
Build (PM)	LOS E · 67.6 s/veh · v/c 1.05
Δ control delay	+28.0 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +28.0s, design year +53.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 75 s cycle, 3 critical phases; g/C NS 0.41 / EW 0.26 (NS left 0.13); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ WB leg	Σ entering
NB approach	—	381	175	556
SB approach	381	—	175	556
WB approach	175	175	—	350
Σ exiting	556	556	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	572	140	713	0.78 → 0.97	27.1 → 47.9	C → D	599
SB	572	65	637	0.78 → 0.87	27.1 → 33.2	C → C	501
EB	0	0	0	0.00 → 0.00	20.7 → 20.7	C → C	0
WB	361	0	361	0.78 → 0.78	37.9 → 37.9	D → D	288

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	140	68%
SB Through	65	32%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

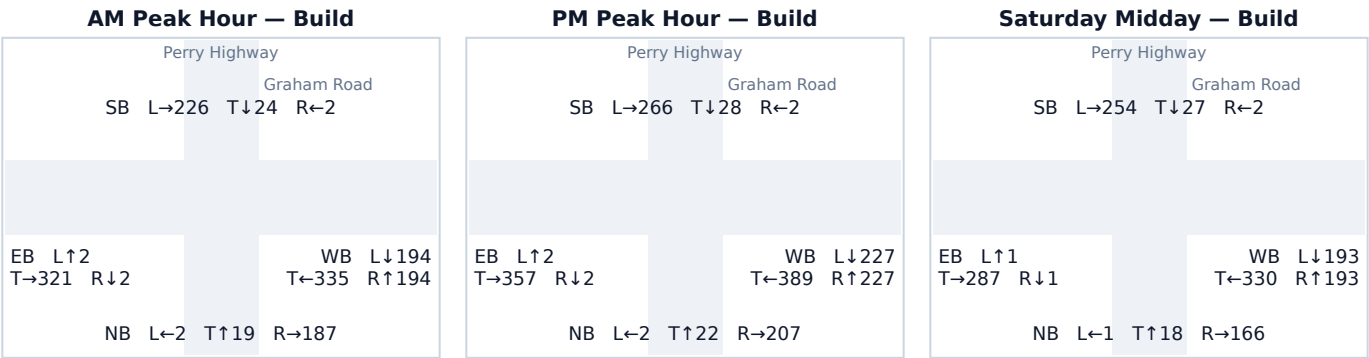
A.2 Perry Highway & Graham Road

Signal ID	pittsburgh-2221
Location	NW Pittsburgh · 1.81 mi from site
Added PM peak trips	135
Existing / No-Build (PM)	LOS D · 50.8 s/veh · v/c 0.99
Build (PM)	LOS E · 75.0 s/veh · v/c 1.08
Δ control delay	+24.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +24.1s, design year +36.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 65 s cycle, 3 critical phases; g/C NS 0.21 / EW 0.40 (EW left 0.15); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 42 iterations, residual 0.4 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	21	2	202	225
SB approach	21	—	2	202	225
EB approach	2	2	—	347	350
WB approach	202	202	347	—	750
Σ exiting	225	225	350	750	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	231	0	231	0.60 → 0.60	29.9 → 29.9	C → C	156
SB	231	65	296	0.60 → 0.77	29.9 → 38.0	C → D	208
EB	361	0	361	0.50 → 0.50	17.0 → 17.0	B → B	201
WB	773	70	843	1.07 → 1.16	72.2 → 107.9	E → F	624

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	70	52%
SB Left	65	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Sewickley Oakmont Road & Lombardy Drive

Signal ID	pittsburgh-2220
Location	NW Pittsburgh · 1.85 mi from site
Added PM peak trips	135
Existing / No-Build (PM)	LOS D · 50.8 s/veh · v/c 0.99
Build (PM)	LOS E · 75.0 s/veh · v/c 1.08
Δ control delay	+24.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +24.1s, design year +36.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 65 s cycle, 3 critical phases; g/C NS 0.21 / EW 0.40 (EW left 0.15); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 42 iterations, residual 0.4 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	21	202	2	225
SB approach	21	—	202	2	225
EB approach	202	202	—	347	750
WB approach	2	2	347	—	350
Σ exiting	225	225	750	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	231	70	301	0.60 → 0.79	29.9 → 39.0	C → D	212
SB	231	65	296	0.60 → 0.77	29.9 → 38.0	C → D	208
EB	773	0	773	1.07 → 1.07	72.2 → 72.2	E → E	572
WB	361	0	361	0.50 → 0.50	17.0 → 17.0	B → B	201

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	70	52%
SB Through	65	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

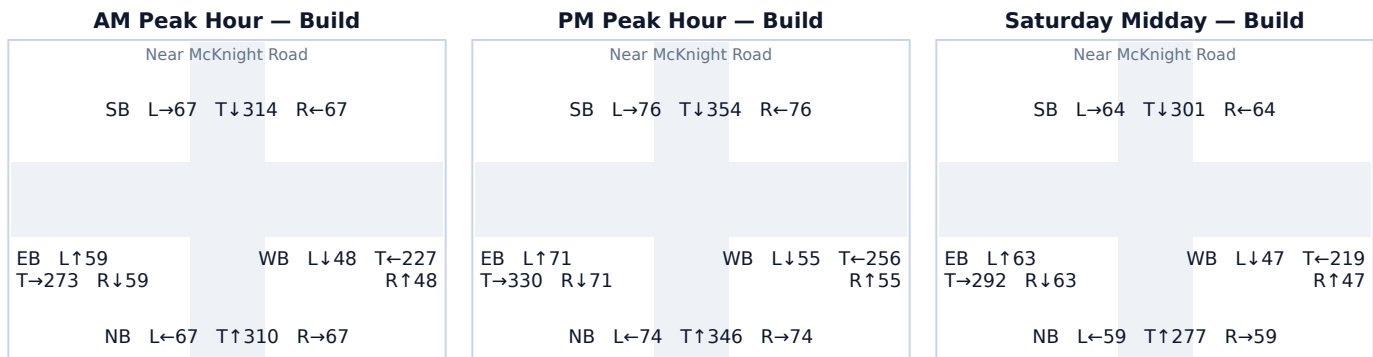
A.4 Near McKnight Road

Signal ID	pittsburgh-2386
Location	NW Pittsburgh · 0.38 mi from site
Added PM peak trips	130
Existing / No-Build (PM)	LOS C · 33.9 s/veh · v/c 0.93
Build (PM)	LOS D · 48.1 s/veh · v/c 1.00
Δ control delay	+14.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +14.2s, design year +30.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.46 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: screening allocation (this junction did not resolve to the road network); turning movements: screening 15/70/15.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	494	0	494	0.60 → 0.60	15.3 → 15.3	B → B	253
SB	472	34	506	0.57 → 0.61	14.8 → 15.7	B → B	262
EB	404	68	472	0.60 → 0.70	19.0 → 21.8	B → C	275
WB	338	28	366	0.50 → 0.54	17.0 → 17.8	B → B	197

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	37	28%
SB Right	34	26%
EB Through	31	24%
WB Through	28	22%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

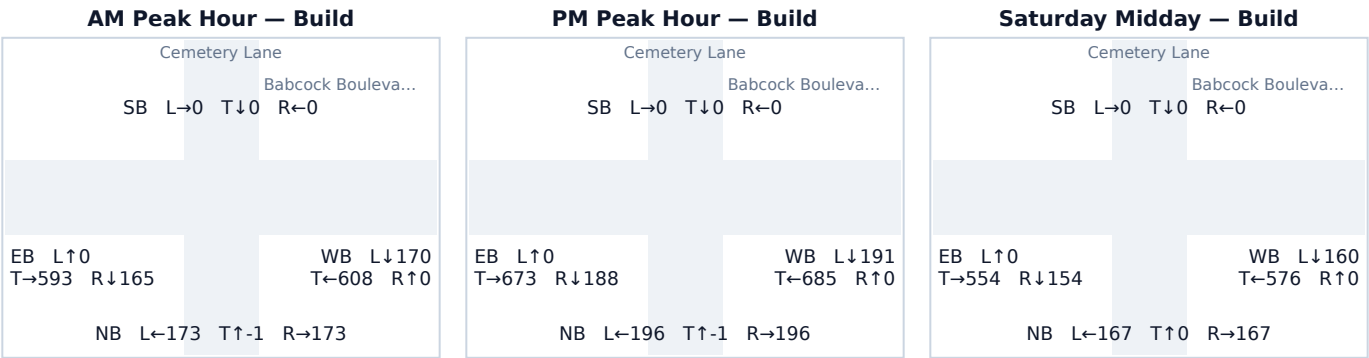
A.5 Cemetery Lane & Babcock Boulevard

Signal ID	pittsburgh-293
Location	NW Pittsburgh · 0.66 mi from site
Added PM peak trips	112
Existing / No-Build (PM)	LOS D · 46.8 s/veh · v/c 0.96
Build (PM)	LOS E · 59.0 s/veh · v/c 1.01
Δ control delay	+12.2 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.2s, design year +23.0s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 115 s cycle, 3 critical phases; g/C NS 0.23 / EW 0.53 (EW left 0.12); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	175	175	350
EB approach	175	—	629	804
WB approach	175	629	—	804
Σ exiting	350	804	804	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	361	30	391	0.87 → 0.95	64.6 → 76.6	E → E	507
SB	0	0	0	0.00 → 0.00	34.2 → 34.2	C → C	0
EB	828	33	861	0.87 → 0.91	35.0 → 38.9	C → D	1,030
WB	828	49	876	0.87 → 0.93	35.0 → 41.2	C → D	1,067

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	49	44%
EB Right	33	29%
NB Left	30	27%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

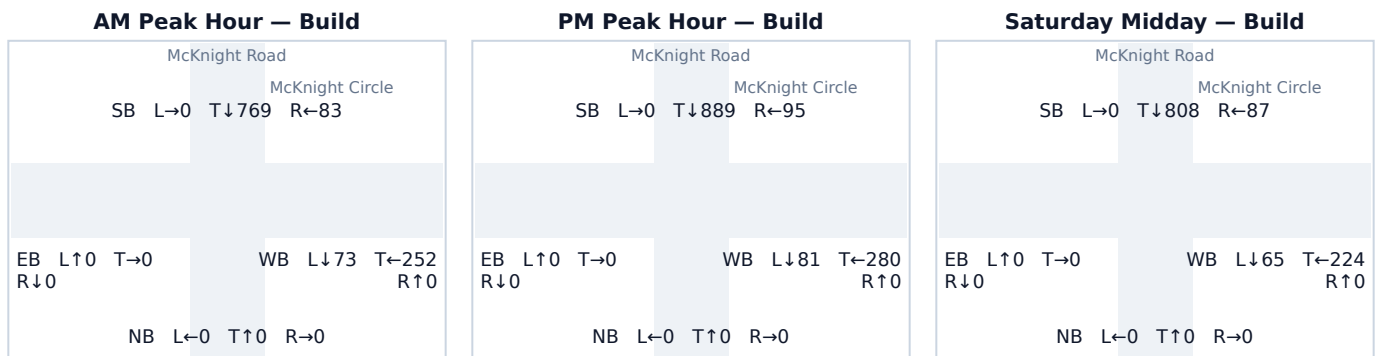
A.6 McKnight Road & McKnight Circle

Signal ID	pittsburgh-2402
Location	NW Pittsburgh · 1.19 mi from site
Added PM peak trips	151
Existing / No-Build (PM)	LOS A · 9.3 s/veh · v/c 0.51
Build (PM)	LOS B · 10.2 s/veh · v/c 0.58
Δ control delay	+0.9 s/veh (LOS change)
Mitigation	No mitigation indicated at screening level (opening year +1.0s, design year +1.9s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.58 / EW 0.25; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	730	—	79	0	809
EB approach	0	0	—	0	0
WB approach	78	0	272	—	350
Σ exiting	809	0	350	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0
SB	834	151	984	0.80 → 0.94	16.1 → 28.3	B → C	625
EB	0	0	0	0.00 → 0.00	16.8 → 16.8	B → B	0
WB	361	0	361	0.80 → 0.80	34.5 → 34.5	C → C	232

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	151	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

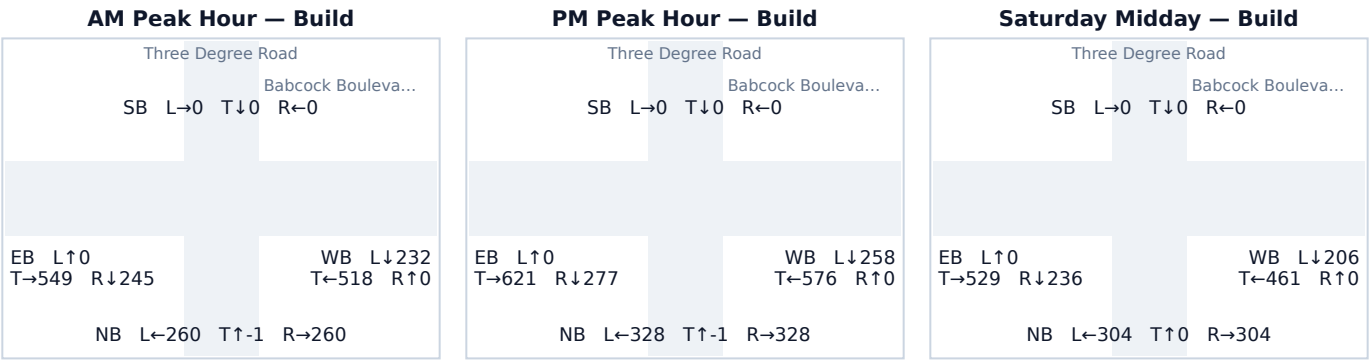
A.7 Three Degree Road & Babcock Boulevard

Signal ID	pittsburgh-285
Location	NW Pittsburgh · 1.68 mi from site
Added PM peak trips	205
Existing / No-Build (PM)	LOS F · 131.4 s/veh · v/c 1.21
Build (PM)	LOS F · 179.8 s/veh · v/c 1.33
Δ control delay	+48.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +48.4s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.89 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	250	250	500
EB approach	250	—	559	809
WB approach	250	559	—	809
Σ exiting	500	809	809	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	515	140	655	0.64 → 0.81	22.9 → 30.0	C → C	584

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	834	65	898	1.03 → 1.11	63.8 → 90.4	E → F	919
WB	834	0	834	1.03 → 1.03	63.8 → 63.8	E → E	853

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Left	70	34%
NB Right	70	34%
EB Right	65	32%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

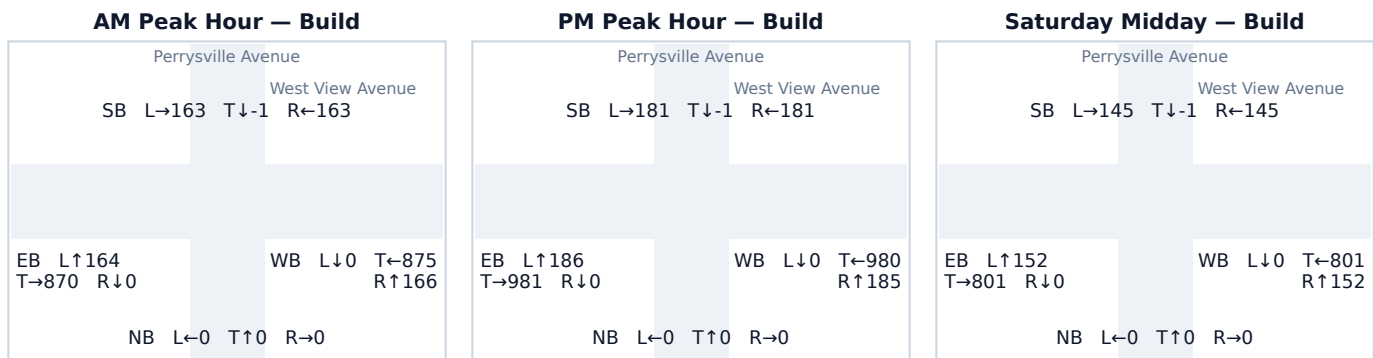
A.8 Perrysville Avenue & West View Avenue

Signal ID	pittsburgh-2208
Location	Central Pittsburgh · 1.78 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS F · 238.7 s/veh · v/c 1.46
Build (PM)	LOS F · 254.1 s/veh · v/c 1.50
Δ control delay	+15.4 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +15.4s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.93 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 2 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	175	175	350
EB approach	175	—	926	1,101
WB approach	175	926	—	1,101
Σ exiting	350	1,101	1,101	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	361	0	361	0.45 → 0.45	18.8 → 18.8	B → B	256
EB	1,134	33	1,167	1.40 → 1.44	212.2 → 229.9	F → F	1,194
WB	1,134	30	1,165	1.40 → 1.44	212.2 → 228.5	F → F	1,191

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	33	52%
WB Through	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

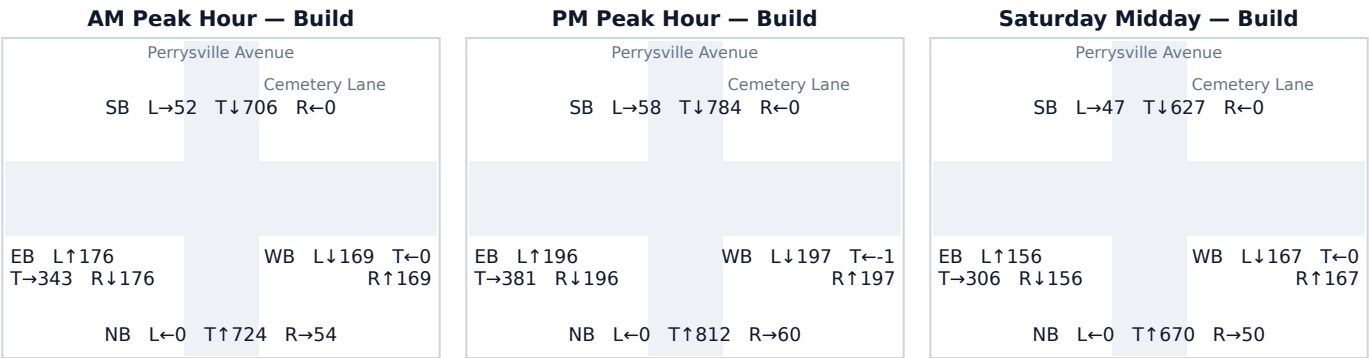
A.9 Perrysville Avenue & Cemetery Lane

Signal ID	pittsburgh-1826
Location	Central Pittsburgh · 1.62 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS F · 285.0 s/veh · v/c 1.57
Build (PM)	LOS F · 300.0 s/veh · v/c 1.60
Δ control delay	+15.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +15.0s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.90 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.1 vph; exits scaled to entries, 27% imbalance). 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	761	0	56	818
SB approach	761	—	0	56	818
EB approach	190	190	—	370	750
WB approach	175	175	0	—	350
Σ exiting	1,126	1,126	0	482	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	842	30	872	1.04 → 1.08	67.0 → 79.0	E → E	892
SB	842	0	842	1.04 → 1.04	67.0 → 67.0	E → E	862
EB	773	0	773	0.95 → 0.95	46.1 → 46.1	D → D	768
WB	361	33	393	0.45 → 0.49	18.8 → 19.5	B → B	285

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	33	52%
NB Right	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

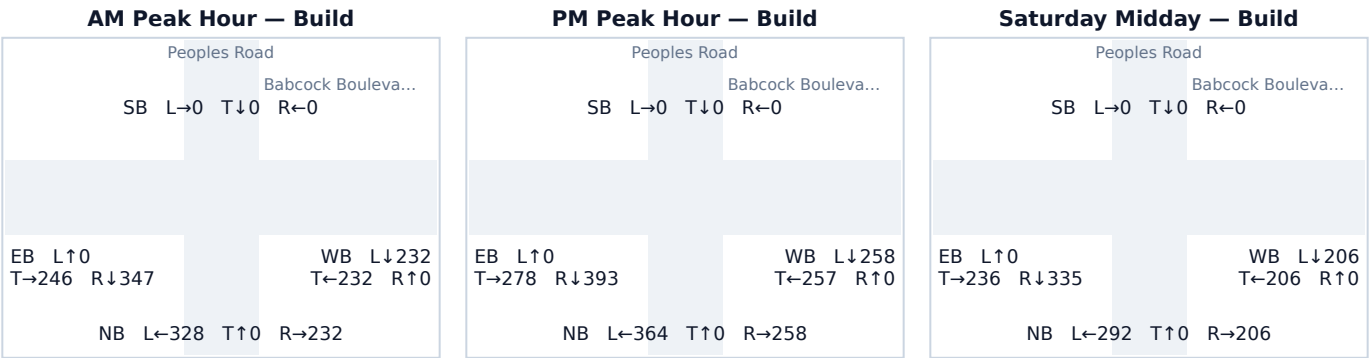
A.10 Peoples Road & Babcock Boulevard

Signal ID	pittsburgh-297
Location	Central Pittsburgh · 1.76 mi from site
Added PM peak trips	49
Existing / No-Build (PM)	LOS F · 147.8 s/veh · v/c 1.21
Build (PM)	LOS F · 161.7 s/veh · v/c 1.25
Δ control delay	+13.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +13.9s, design year +0.0s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 120 s cycle, 3 critical phases; g/C NS 0.36 / EW 0.36 (EW left 0.15); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.2 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	354	250	604
EB approach	354	—	250	604
WB approach	250	250	—	500
Σ exiting	604	604	500	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	622	0	622	0.95 → 0.95	62.8 → 62.8	E → E	834
SB	0	0	0	0.00 → 0.00	24.4 → 24.4	C → C	0
EB	622	49	671	0.95 → 1.03	62.8 → 80.6	E → F	917
WB	515	0	515	0.79 → 0.79	43.6 → 43.6	D → D	633

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	49	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

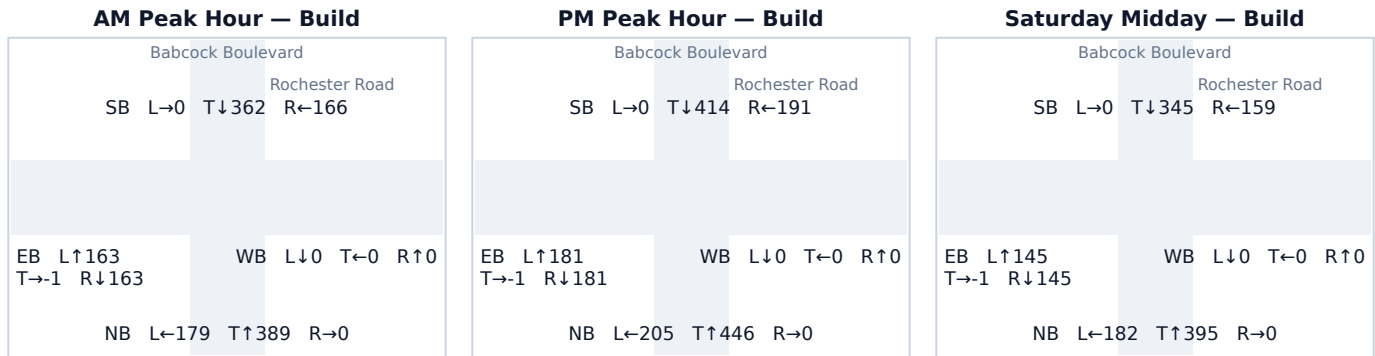
A.11 Babcock Boulevard & Rochester Road

Signal ID	pittsburgh-294
Location	NW Pittsburgh · 0.40 mi from site
Added PM peak trips	112
Existing / No-Build (PM)	LOS D · 39.6 s/veh · v/c 0.92
Build (PM)	LOS D · 52.4 s/veh · v/c 0.99
Δ control delay	+12.8 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +12.9s, design year +28.9s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 75 s cycle, 3 critical phases; g/C NS 0.41 / EW 0.26 (NS left 0.13); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	381	175	556
SB approach	381	—	175	556
EB approach	175	175	—	350
Σ exiting	556	556	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	572	79	651	0.78 → 0.88	27.1 → 35.1	C → D	518
SB	572	33	605	0.78 → 0.82	27.1 → 29.8	C → C	463
EB	361	0	361	0.78 → 0.78	37.9 → 37.9	D → D	288
WB	0	0	0	0.00 → 0.00	20.7 → 20.7	C → C	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	79	71%
SB Through	33	29%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

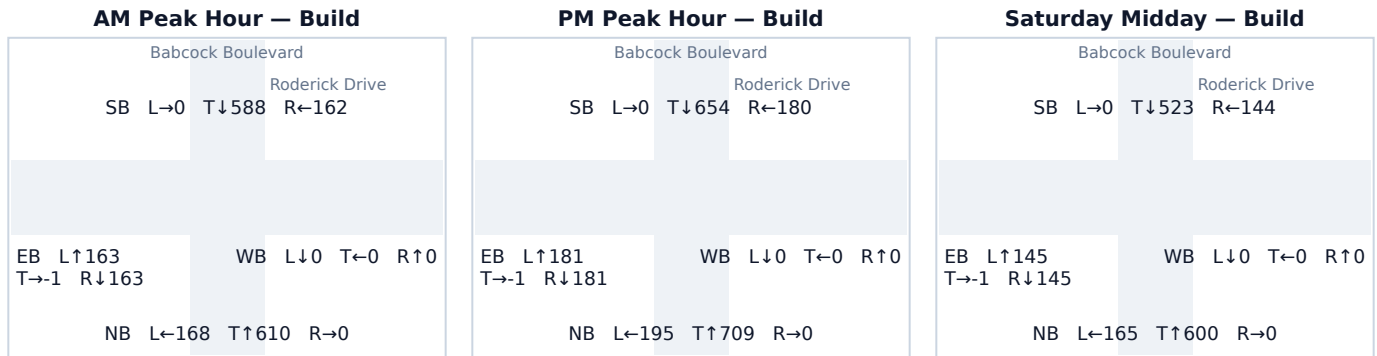
A.12 Babcock Boulevard & Roderick Drive

Signal ID	pittsburgh-286
Location	NW Pittsburgh · 1.95 mi from site
Added PM peak trips	70
Existing / No-Build (PM)	LOS D · 46.9 s/veh · v/c 0.96
Build (PM)	LOS D · 54.2 s/veh · v/c 0.99
Δ control delay	+7.3 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +7.3s, design year +14.3s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 120 s cycle, 3 critical phases; g/C NS 0.53 / EW 0.23 (NS left 0.12); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	634	175	809
SB approach	634	—	175	809
EB approach	175	175	—	350
Σ exiting	809	809	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	834	70	904	0.87 → 0.95	35.5 → 45.2	D → D	1,171
SB	834	0	834	0.87 → 0.87	35.5 → 35.5	D → D	1,002
EB	361	0	361	0.87 → 0.87	66.2 → 66.2	E → E	478
WB	0	0	0	0.00 → 0.00	35.6 → 35.6	D → D	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	70	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

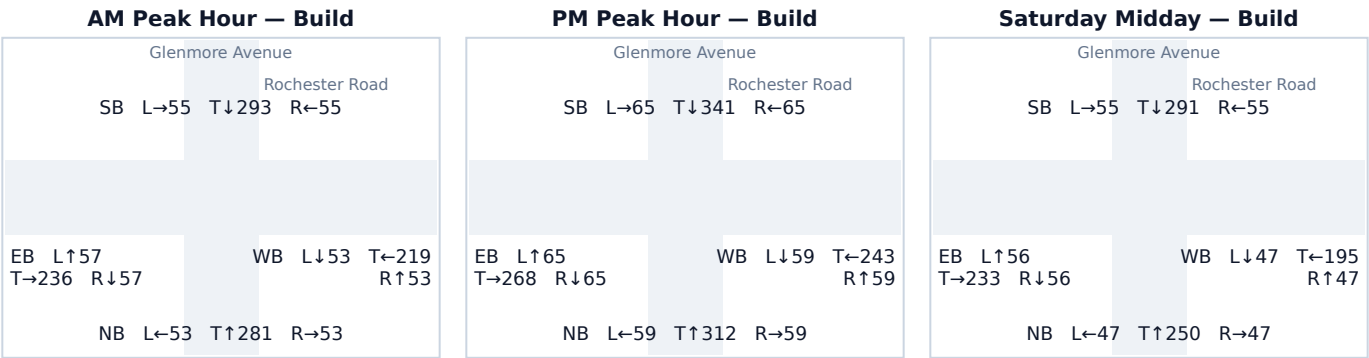
A.13 Glenmore Avenue & Rochester Road

Signal ID	pittsburgh-109
Location	NW Pittsburgh · 0.71 mi from site
Added PM peak trips	78
Existing / No-Build (PM)	LOS C · 27.2 s/veh · v/c 0.87
Build (PM)	LOS C · 31.9 s/veh · v/c 0.92
Δ control delay	+4.7 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +4.7s, design year +17.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.38; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	303	57	57	418
SB approach	303	—	57	57	418
EB approach	57	57	—	236	350
WB approach	57	57	236	—	350
Σ exiting	417	417	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	430	0	430	0.53 → 0.53	14.2 → 14.2	B → B	212
SB	430	41	471	0.53 → 0.58	14.2 → 15.1	B → B	240
EB	361	37	398	0.53 → 0.58	17.3 → 18.4	B → B	218
WB	361	0	361	0.53 → 0.53	17.3 → 17.3	B → B	192

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	41	53%
EB Left	37	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

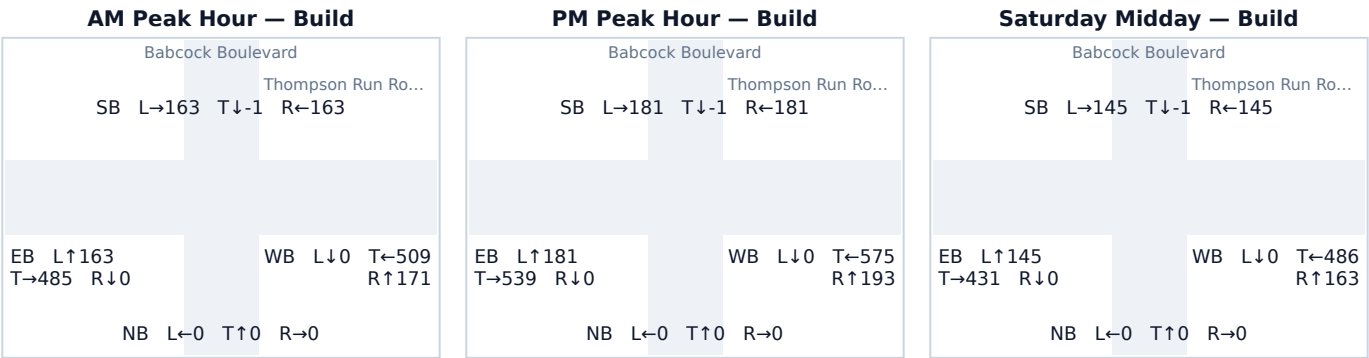
A.14 Babcock Boulevard & Thompson Run Road

Signal ID	pittsburgh-292
Location	NE Pittsburgh · 1.39 mi from site
Added PM peak trips	49
Existing / No-Build (PM)	LOS D · 41.8 s/veh · v/c 0.94
Build (PM)	LOS D · 46.3 s/veh · v/c 0.96
Δ control delay	+4.5 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +4.5s, design year +10.8s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 95 s cycle, 3 critical phases; g/C NS 0.24 / EW 0.48 (EW left 0.12); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	175	175	350
EB approach	175	—	524	699
WB approach	175	524	—	699
Σ exiting	350	699	699	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	27.4 → 27.4	C → C	0
SB	361	0	361	0.83 → 0.83	51.0 → 51.0	D → D	373
EB	720	0	720	0.83 → 0.83	30.5 → 30.5	C → C	678
WB	720	49	768	0.83 → 0.89	30.5 → 35.4	C → D	757

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	49	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

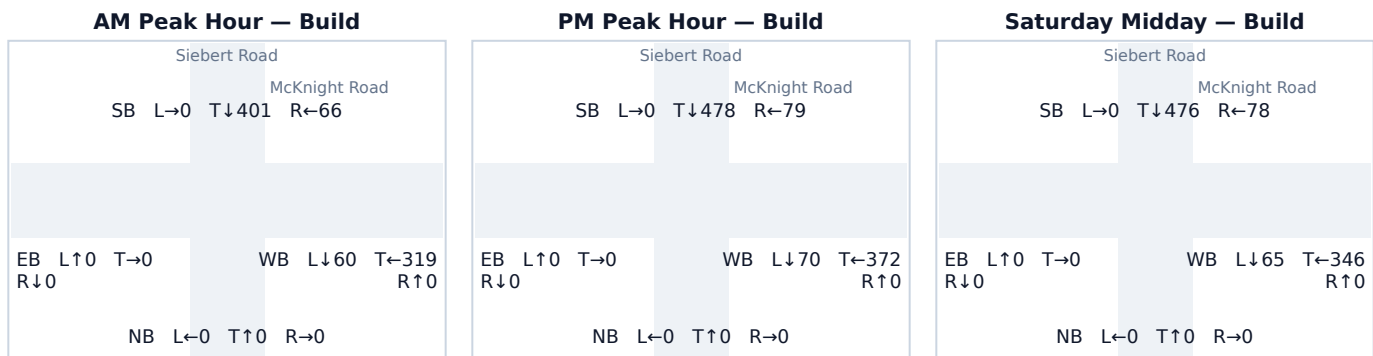
A.15 Siebert Road & McKnight Road

Signal ID	pittsburgh-2382
Location	NW Pittsburgh · 0.41 mi from site
Added PM peak trips	232
Existing / No-Build (PM)	LOS B · 13.3 s/veh · v/c 0.43
Build (PM)	LOS B · 15.4 s/veh · v/c 0.57
Δ control delay	+2.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +2.1s, design year +3.4s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 5 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	339	—	56	0	395
EB approach	0	0	—	0	0
WB approach	56	0	294	—	350
Σ exiting	394	0	350	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	9.4 → 9.4	A → A	0
SB	406	151	557	0.51 → 0.70	14.4 → 18.7	B → B	310
EB	0	0	0	0.00 → 0.00	11.1 → 11.1	B → B	0
WB	361	81	442	0.51 → 0.63	16.5 → 18.9	B → B	245

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Right	151	65%
WB Through	81	35%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

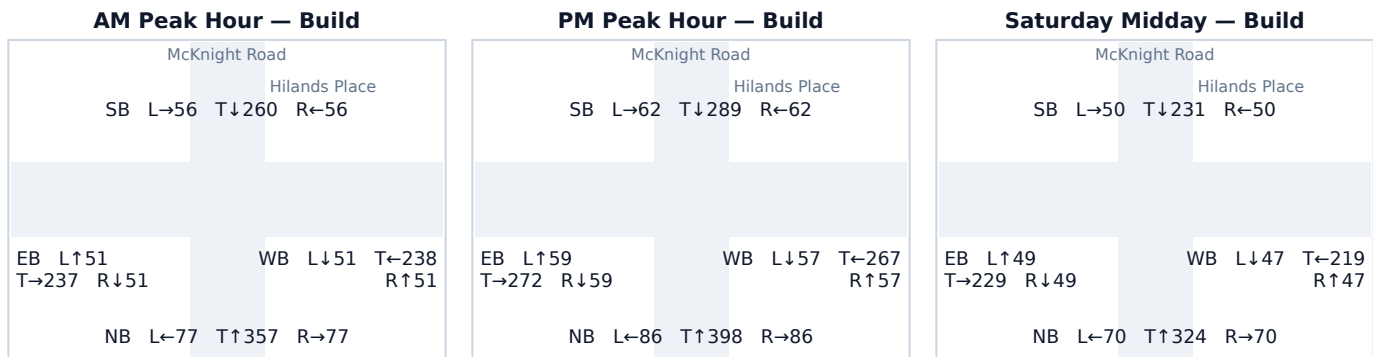
A.16 McKnight Road & Hilands Place

Signal ID	pittsburgh-2397
Location	NW Pittsburgh · 0.58 mi from site
Added PM peak trips	47
Existing / No-Build (PM)	LOS C · 22.8 s/veh · v/c 0.85
Build (PM)	LOS C · 24.7 s/veh · v/c 0.87
Δ control delay	+1.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +1.9s, design year +9.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.50 / EW 0.33; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: screening allocation (this junction did not resolve to the road network); turning movements: screening 15/70/15.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	560	11	570	0.62 → 0.63	14.0 → 14.2	B → B	286
SB	413	0	413	0.46 → 0.46	11.3 → 11.3	B → B	184
EB	365	24	390	0.61 → 0.65	21.5 → 22.7	C → C	229
WB	369	12	381	0.62 → 0.64	21.7 → 22.2	C → C	222

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	13	28%
WB Through	12	26%
EB Right	11	23%
NB Left	11	23%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

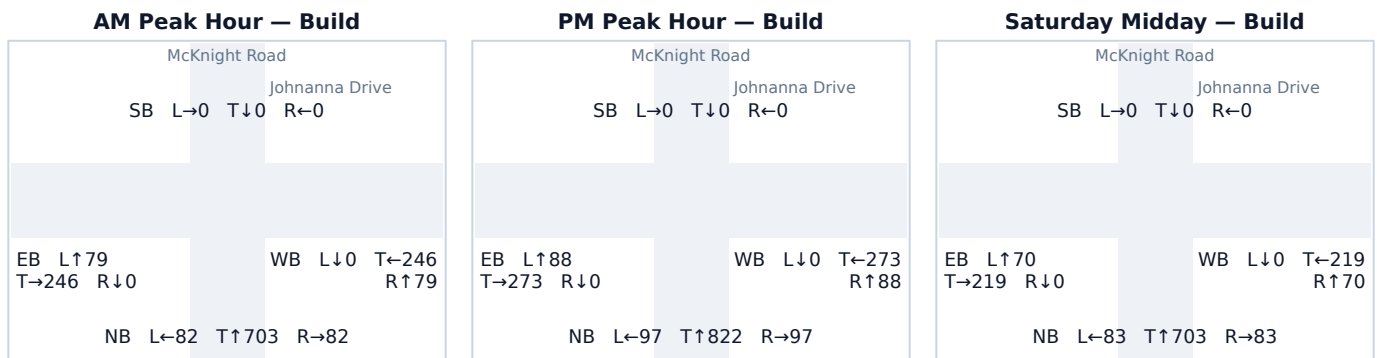
A.17 McKnight Road & Johnanna Drive

Signal ID	pittsburgh-2913
Location	NW Pittsburgh · 1.45 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS B · 11.9 s/veh · v/c 0.67
Build (PM)	LOS B · 12.9 s/veh · v/c 0.70
Δ control delay	+1.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.9s, design year +4.6s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 70 s cycle, 2 critical phases; g/C NS 0.62 / EW 0.24; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 6 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	725	85	85	896
SB approach	0	—	0	0	0
EB approach	0	85	—	265	350
WB approach	0	85	265	—	350
Σ exiting	0	895	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	923	93	1,016	0.83 → 0.92	17.9 → 25.0	B → C	718
SB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0
EB	361	0	361	0.83 → 0.83	42.0 → 42.0	D → D	275
WB	361	0	361	0.83 → 0.83	42.0 → 42.0	D → D	275

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

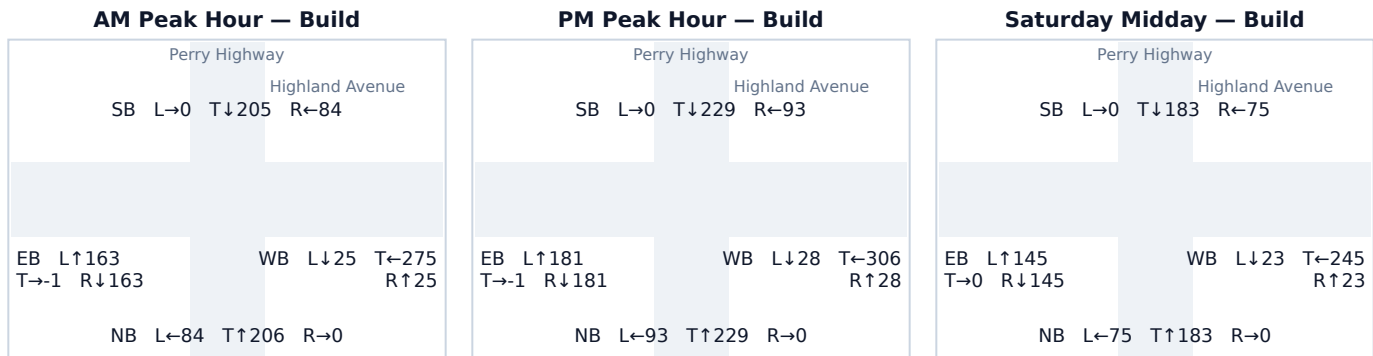
A.18 Perry Highway & Highland Avenue

Signal ID	pittsburgh-2229
Location	NW Pittsburgh · 1.14 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS F · 108.5 s/veh · v/c 1.15
Build (PM)	LOS F · 109.4 s/veh · v/c 1.15
Δ control delay	+0.9 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +0.9s, design year +1.1s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 3 critical phases; g/C NS 0.29 / EW 0.30 (EW left 0.17); permissive NS / protected EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.4 vph; exits scaled to entries, 26% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	222	90	0	312
SB approach	222	—	90	0	312
EB approach	175	175	—	0	350
WB approach	28	28	295	—	350
Σ exiting	424	424	476	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	321	0	322	0.62 → 0.62	24.1 → 24.2	C → C	192
SB	321	0	322	0.62 → 0.62	24.1 → 24.2	C → C	192
EB	361	1	361	0.68 → 0.68	25.3 → 25.3	C → C	219
WB	361	2	362	0.68 → 0.68	25.3 → 25.4	C → C	219

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	33%
EB Through	1	33%
WB Left	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

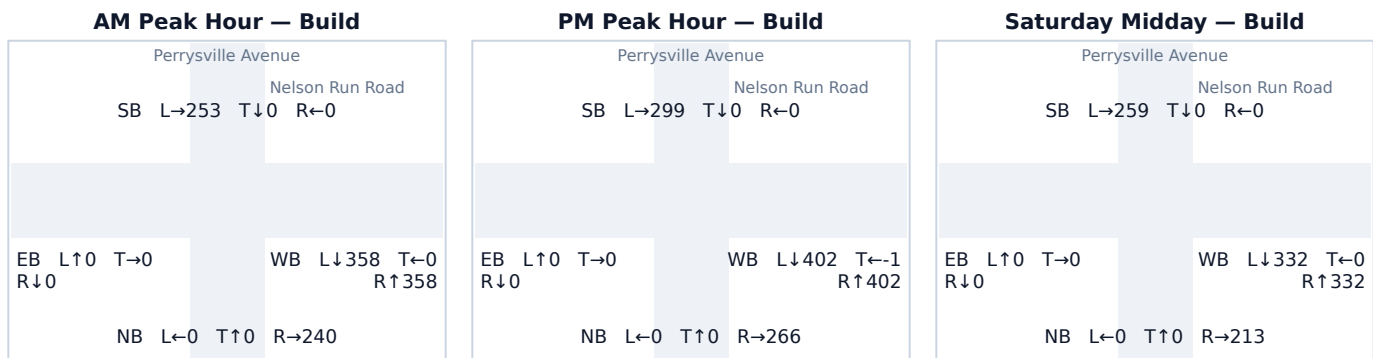
A.19 Perrysville Avenue & Nelson Run Road

Signal ID	pittsburgh-2212
Location	Central Pittsburgh · 1.89 mi from site
Added PM peak trips	63
Existing / No-Build (PM)	LOS C · 22.4 s/veh · v/c 0.64
Build (PM)	LOS C · 23.3 s/veh · v/c 0.67
Δ control delay	+0.9 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +0.9s, design year +3.2s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 105 s cycle, 3 critical phases; g/C NS 0.17 / EW 0.51 (NS left 0.17); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: did not balance within 50 iterations (residual 233.0 vph; rows held exact).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ WB leg	Σ entering
NB approach	—	0	259	259
SB approach	0	—	259	259
WB approach	375	375	—	750
Σ exiting	375	375	517	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	266	0	266	0.85 → 0.85	65.4 → 65.4	E → E	310

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
SB	266	33	299	0.85 → 0.95	65.4 → 82.3	E → F	356
EB	0	0	0	0.00 → 0.00	12.7 → 12.7	B → B	0
WB	773	30	803	0.85 → 0.88	31.8 → 34.8	C → C	859

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Left	33	52%
WB Right	30	48%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.20 Babcock Boulevard & Browns Lane

Signal ID	pittsburgh-1393
Location	NW Pittsburgh · 1.90 mi from site
Added PM peak trips	70
Existing / No-Build (PM)	LOS B · 11.8 s/veh · v/c 0.69
Build (PM)	LOS B · 12.5 s/veh · v/c 0.72
Δ control delay	+0.7 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.8s, design year +4.7s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 1 extra leg not carried (4×4 matrix).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	809	809
WB approach	809	—	809
Σ exiting	809	809	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	834	70	904	0.77 → 0.83	14.1 → 17.1	B → B	496
WB	834	0	834	0.77 → 0.77	14.1 → 14.1	B → B	424

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	70	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

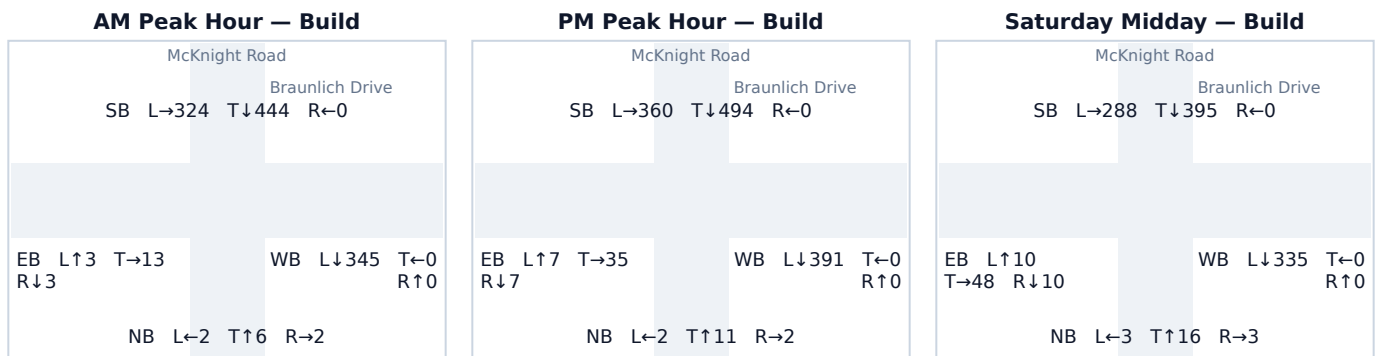
A.21 McKnight Road & Braunlich Drive

Signal ID	pittsburgh-2393
Location	NW Pittsburgh · 0.44 mi from site
Added PM peak trips	94
Existing / No-Build (PM)	LOS A · 9.4 s/veh · v/c 0.51
Build (PM)	LOS A · 10.0 s/veh · v/c 0.55
Δ control delay	+0.6 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +1.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 65 s cycle, 2 critical phases; g/C NS 0.59 / EW 0.25; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ WB leg	Σ entering
NB approach	—	0	0	0
SB approach	479	—	350	829
WB approach	350	0	—	350
Σ exiting	829	0	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	14	15	0.00 → 0.01	5.3 → 5.4	A → A	4
SB	854	0	854	0.80 → 0.80	16.3 → 16.3	B → B	490
EB	0	49	49	0.00 → 0.11	18.2 → 19.2	B → B	28
WB	361	31	391	0.80 → 0.86	36.4 → 42.6	D → D	279

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	33	35%
WB Through	31	33%
EB Right	16	17%
NB Left	14	15%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

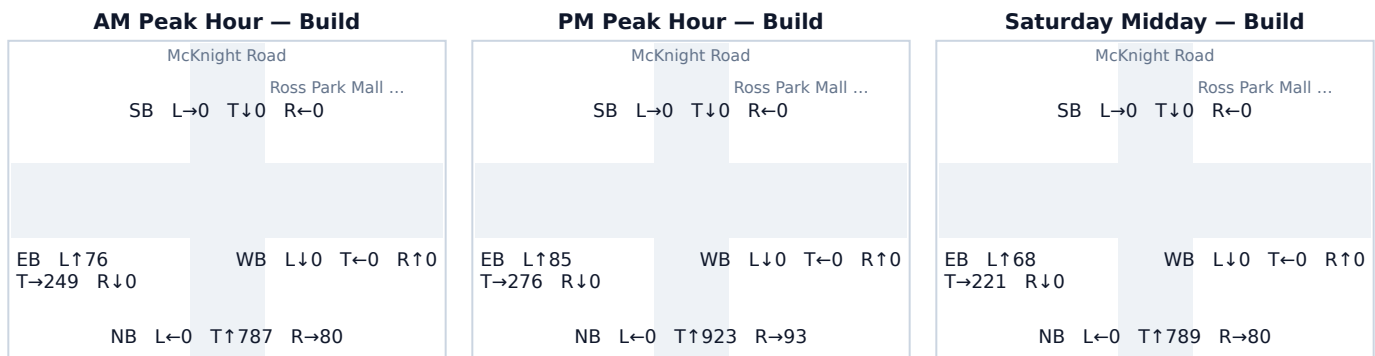
A.22 McKnight Road & Ross Park Mall Drive

Signal ID	pittsburgh-2404
Location	NW Pittsburgh · 1.17 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS A · 9.3 s/veh · v/c 0.52
Build (PM)	LOS A · 9.9 s/veh · v/c 0.56
Δ control delay	+0.6 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +1.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 70 s cycle, 2 critical phases; g/C NS 0.62 / EW 0.24; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	813	0	83	896
SB approach	0	—	0	0	0
EB approach	0	82	—	268	350
WB approach	0	0	0	—	0
Σ exiting	0	895	0	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	923	93	1,016	0.83 → 0.92	17.9 → 25.0	B → C	718
SB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0
EB	361	0	361	0.83 → 0.83	42.0 → 42.0	D → D	275
WB	0	0	0	0.00 → 0.00	20.2 → 20.2	C → C	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

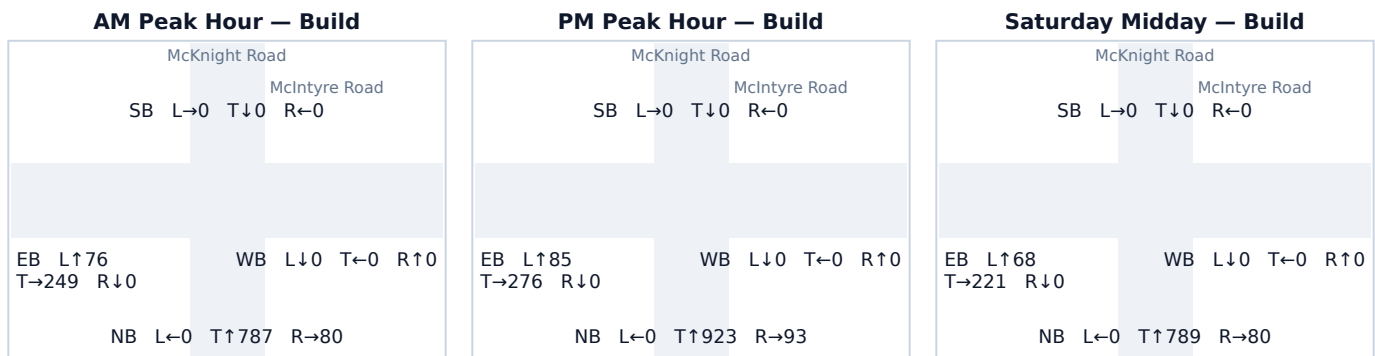
A.23 McKnight Road & McIntyre Road

Signal ID	pittsburgh-2415
Location	NW Pittsburgh · 1.78 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS A · 9.3 s/veh · v/c 0.52
Build (PM)	LOS A · 9.9 s/veh · v/c 0.56
Δ control delay	+0.6 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +1.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 70 s cycle, 2 critical phases; g/C NS 0.62 / EW 0.24; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	813	0	83	896
SB approach	0	—	0	0	0
EB approach	0	82	—	268	350
WB approach	0	0	0	—	0
Σ exiting	0	895	0	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	923	93	1,016	0.83 → 0.92	17.9 → 25.0	B → C	718
SB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0
EB	361	0	361	0.83 → 0.83	42.0 → 42.0	D → D	275
WB	0	0	0	0.00 → 0.00	20.2 → 20.2	C → C	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

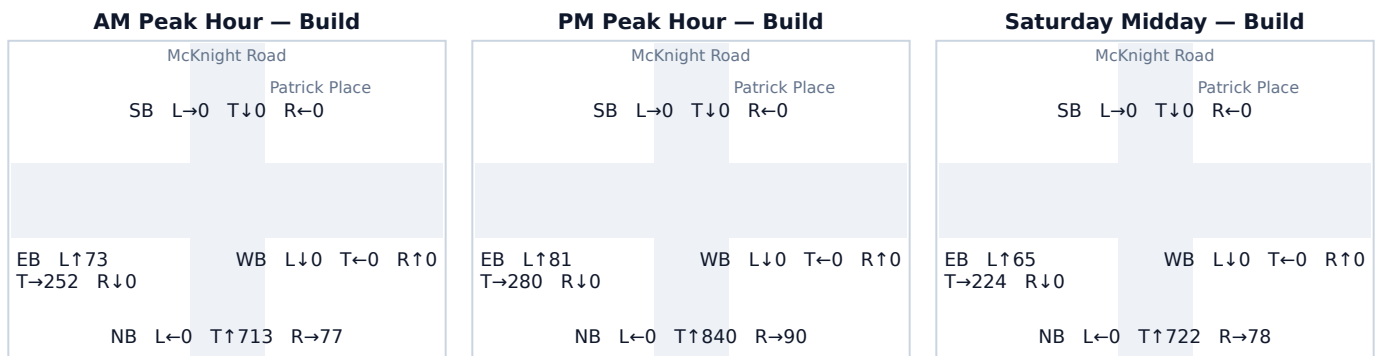
A.24 McKnight Road & Patrick Place

Signal ID	pittsburgh-2407
Location	NW Pittsburgh · 0.87 mi from site
Added PM peak trips	93
Existing / No-Build (PM)	LOS A · 9.3 s/veh · v/c 0.51
Build (PM)	LOS A · 9.8 s/veh · v/c 0.55
Δ control delay	+0.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +1.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.58 / EW 0.25; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 7 iterations, residual 0.3 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	733	0	79	812
SB approach	0	—	0	0	0
EB approach	0	79	—	271	350
WB approach	0	0	0	—	0
Σ exiting	0	812	0	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	837	93	930	0.80 → 0.89	16.1 → 21.9	B → C	553
SB	0	0	0	0.00 → 0.00	5.2 → 5.2	A → A	0
EB	361	0	361	0.80 → 0.80	34.7 → 34.7	C → C	232
WB	0	0	0	0.00 → 0.00	16.8 → 16.8	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	93	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

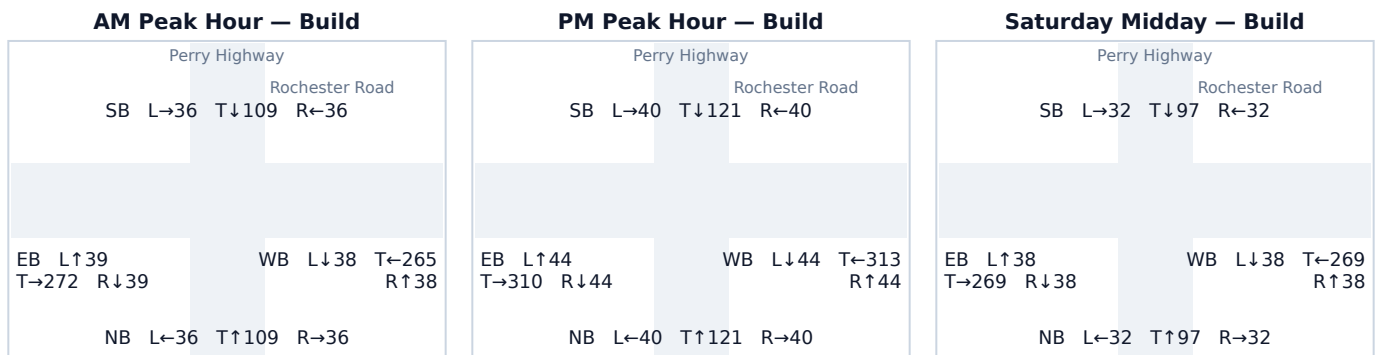
A.25 Perry Highway & Rochester Road

Signal ID	pittsburgh-2228
Location	NW Pittsburgh · 1.19 mi from site
Added PM peak trips	78
Existing / No-Build (PM)	LOS B · 11.1 s/veh · v/c 0.52
Build (PM)	LOS B · 11.6 s/veh · v/c 0.56
Δ control delay	+0.5 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.6s, design year +1.2s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.30 / EW 0.54; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	118	38	38	195
SB approach	118	—	38	38	195
EB approach	39	39	—	273	350
WB approach	39	39	273	—	350
Σ exiting	195	195	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	201	0	201	0.37 → 0.37	18.6 → 18.6	B → B	109
SB	201	0	201	0.37 → 0.37	18.6 → 18.6	B → B	109
EB	361	37	398	0.37 → 0.41	9.2 → 9.6	A → A	163
WB	361	41	401	0.37 → 0.42	9.2 → 9.7	A → A	165

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	41	53%
EB Through	37	47%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

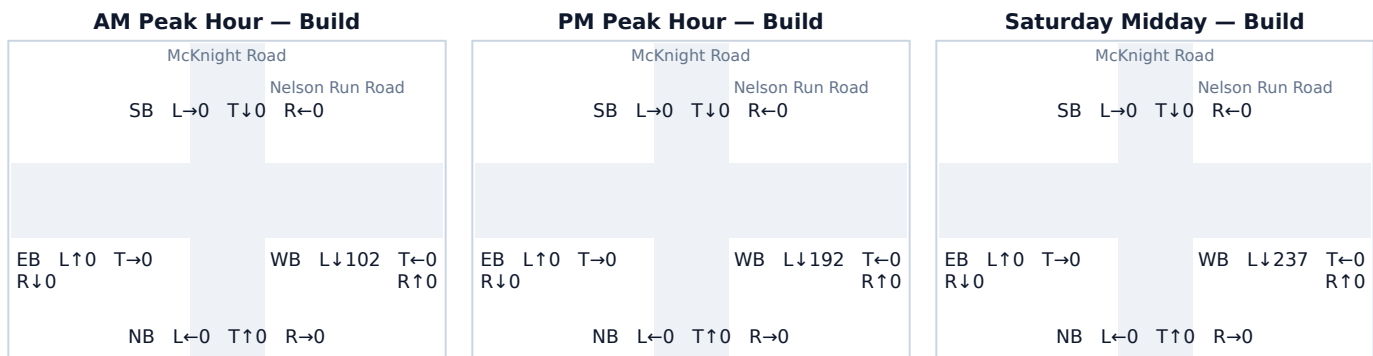
A.26 McKnight Road & Nelson Run Road

Signal ID	pittsburgh-2379
Location	Central Pittsburgh · 1.29 mi from site
Added PM peak trips	140
Existing / No-Build (PM)	LOS A · 4.8 s/veh · v/c 0.02
Build (PM)	LOS A · 5.1 s/veh · v/c 0.08
Δ control delay	+0.3 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.3s, design year +0.3s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ WB leg	Σ entering
NB approach	—	0	0
WB approach	50	—	50
Σ exiting	50	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	0	0	0	0.00 → 0.00	4.7 → 4.7	A → A	0
WB	52	140	192	0.05 → 0.18	5.0 → 5.7	A → A	59

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	140	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

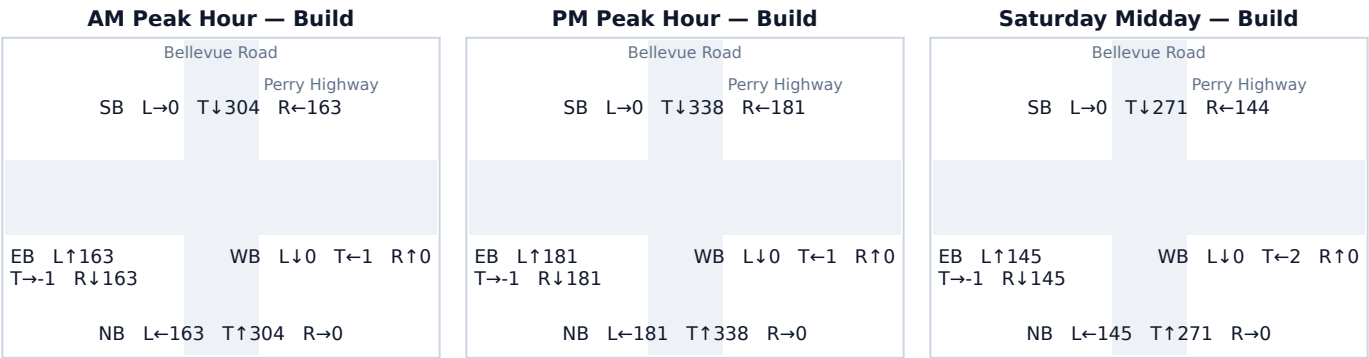
A.27 Bellevue Road & Perry Highway

Signal ID	pittsburgh-1260
Location	NW Pittsburgh · 1.18 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS D · 40.6 s/veh · v/c 0.92
Build (PM)	LOS D · 40.8 s/veh · v/c 0.92
Δ control delay	+0.2 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.2s, design year +0.6s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 70 s cycle, 3 critical phases; g/C NS 0.38 / EW 0.26 (NS left 0.14); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.0 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	329	175	504
SB approach	329	—	175	504
EB approach	175	175	—	350
Σ exiting	504	504	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	519	0	519	0.76 → 0.76	26.7 → 26.8	C → C	363
SB	519	0	519	0.76 → 0.76	26.7 → 26.7	C → C	363
EB	361	1	361	0.76 → 0.76	34.6 → 34.7	C → C	267
WB	0	1	1	0.00 → 0.00	19.0 → 19.0	B → B	1

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

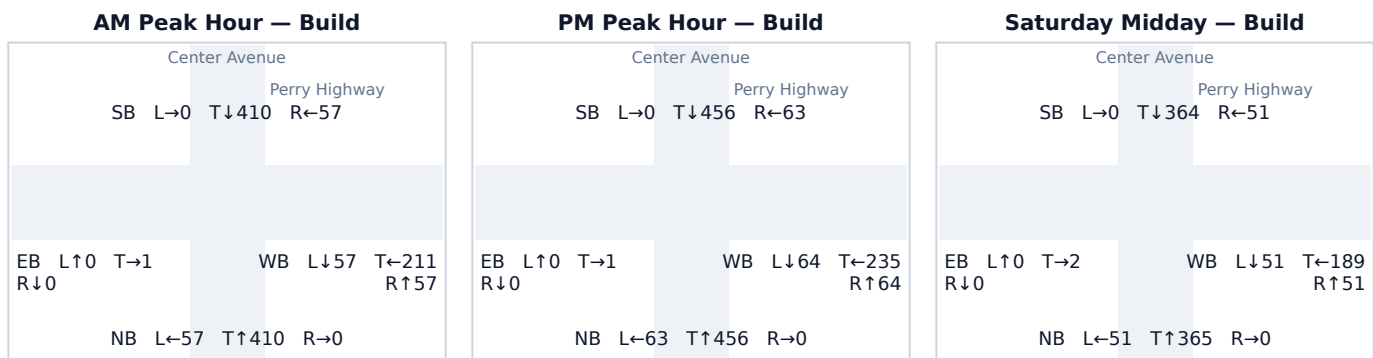
A.28 Center Avenue & Perry Highway

Signal ID	pittsburgh-2217
Location	NW Pittsburgh · 1.08 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 16.7 s/veh · v/c 0.71
Build (PM)	LOS B · 16.8 s/veh · v/c 0.71
Δ control delay	+0.1 s/veh
Mitigation	Minor: signal-timing optimization (shift 3–5s of green to the critical phase) is sufficient. No geometric change required. Governing horizon: design year (opening year +0.1s, design year +0.4s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 5 iterations, residual 0.1 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	442	62	0	504
SB approach	442	—	62	0	504
EB approach	0	0	—	0	0
WB approach	62	62	227	—	350
Σ exiting	504	504	350	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	519	1	519	0.59 → 0.59	13.7 → 13.7	B → B	255
SB	519	0	519	0.59 → 0.59	13.7 → 13.7	B → B	255
EB	0	1	1	0.00 → 0.00	13.0 → 13.0	B → B	1
WB	361	2	363	0.59 → 0.59	20.3 → 20.4	C → C	206

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	25%
EB Through	1	25%
WB Left	1	25%
NB Right	1	25%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

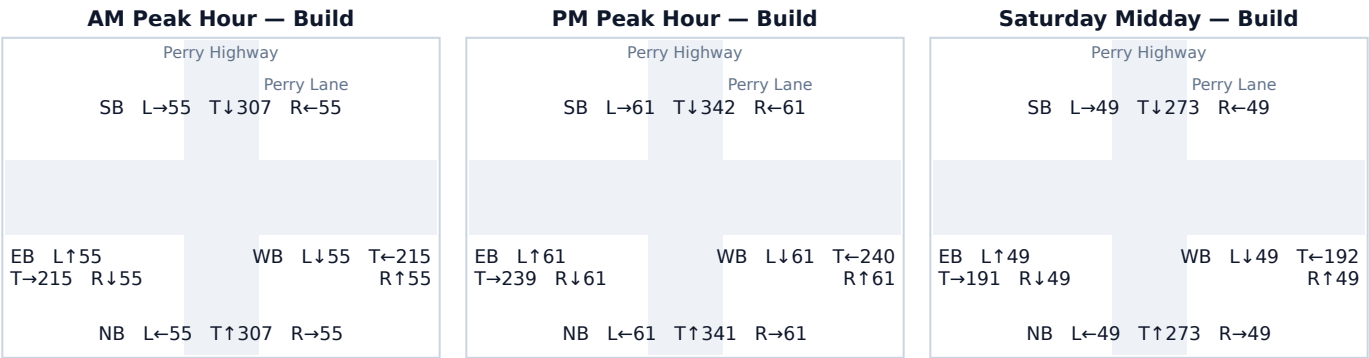
A.29 Perry Highway & Perry Lane

Signal ID	pittsburgh-2227
Location	NW Pittsburgh · 1.21 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS C · 27.0 s/veh · v/c 0.88
Build (PM)	LOS C · 27.1 s/veh · v/c 0.88
Δ control delay	+0.1 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.1s, design year +0.5s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.47 / EW 0.36; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.1 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	331	59	59	449
SB approach	331	—	59	59	449
EB approach	59	59	—	232	350
WB approach	59	59	232	—	350
Σ exiting	449	449	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	463	0	463	0.55 → 0.55	14.0 → 14.0	B → B	228
SB	463	0	464	0.55 → 0.55	14.0 → 14.0	B → B	228
EB	361	1	361	0.55 → 0.55	18.4 → 18.4	B → B	197
WB	361	1	362	0.55 → 0.55	18.4 → 18.5	B → B	198

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

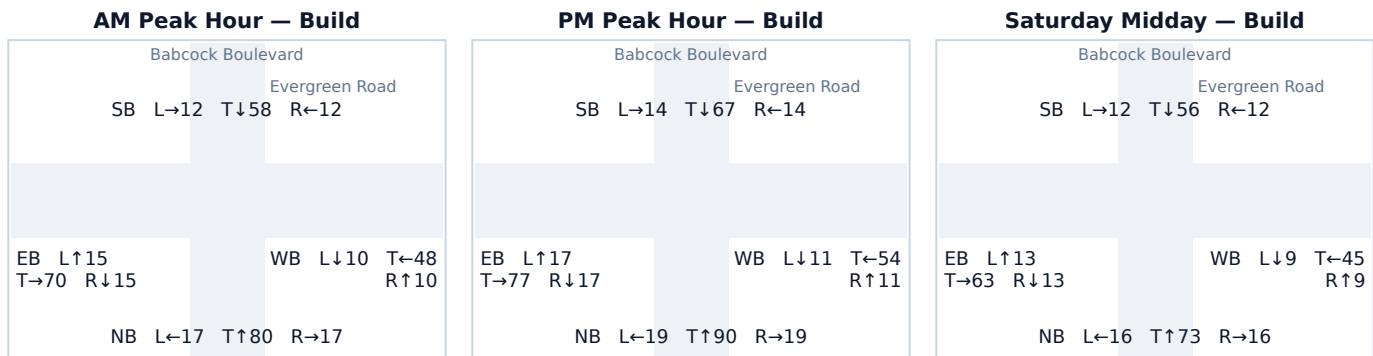
A.30 Babcock Boulevard & Evergreen Road

Signal ID	pittsburgh-2400
Location	NW Pittsburgh · 0.85 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS B · 11.1 s/veh · v/c 0.23
Build (PM)	LOS B · 11.2 s/veh · v/c 0.23
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.1s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.39; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: screening allocation (this junction did not resolve to the road network); turning movements: screening 15/70/15.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	124	4	128	0.16 → 0.16	10.5 → 10.6	B → B	53
SB	89	6	95	0.11 → 0.12	10.2 → 10.2	B → B	39
EB	111	0	111	0.16 → 0.16	12.2 → 12.2	B → B	49
WB	74	2	76	0.10 → 0.11	11.8 → 11.8	B → B	33

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	4	33%
NB Through	4	33%
SB Left	2	17%
WB Right	2	17%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

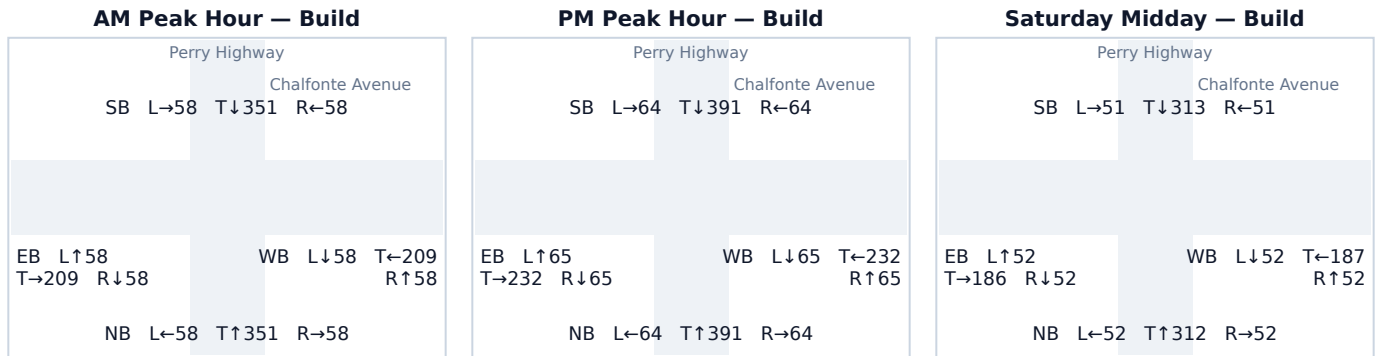
A.31 Perry Highway & Chalfonte Avenue

Signal ID	pittsburgh-813
Location	NW Pittsburgh · 1.12 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS C · 27.3 s/veh · v/c 0.89
Build (PM)	LOS C · 27.4 s/veh · v/c 0.90
Δ control delay	+0.1 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.2s, design year +0.7s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.49 / EW 0.34; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 3 iterations, residual 0.2 vph).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	378	63	63	504
SB approach	378	—	63	63	504
EB approach	63	63	—	225	350
WB approach	63	63	225	—	350
Σ exiting	503	503	350	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	519	0	519	0.59 → 0.59	13.7 → 13.7	B → B	255
SB	519	0	519	0.59 → 0.59	13.7 → 13.7	B → B	255
EB	361	1	362	0.59 → 0.59	20.3 → 20.3	C → C	205
WB	361	2	362	0.59 → 0.59	20.3 → 20.4	C → C	205

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	33%
EB Through	1	33%
WB Left	1	33%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.32 Perry Highway & West View Park Drive

Signal ID	pittsburgh-665
Location	NW Pittsburgh · 1.07 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS A · 7.7 s/veh · v/c 0.43
Build (PM)	LOS A · 7.7 s/veh · v/c 0.43
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 1 extra leg not carried (4×4 matrix).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	504	504
WB approach	504	—	504
Σ exiting	504	504	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	1	1	0.00 → 0.00	17.7 → 17.8	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	519	1	520	0.48 → 0.48	8.2 → 8.2	A → A	200
WB	519	2	521	0.48 → 0.48	8.2 → 8.2	A → A	200

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	25%
EB Through	1	25%
WB Left	1	25%
NB Right	1	25%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

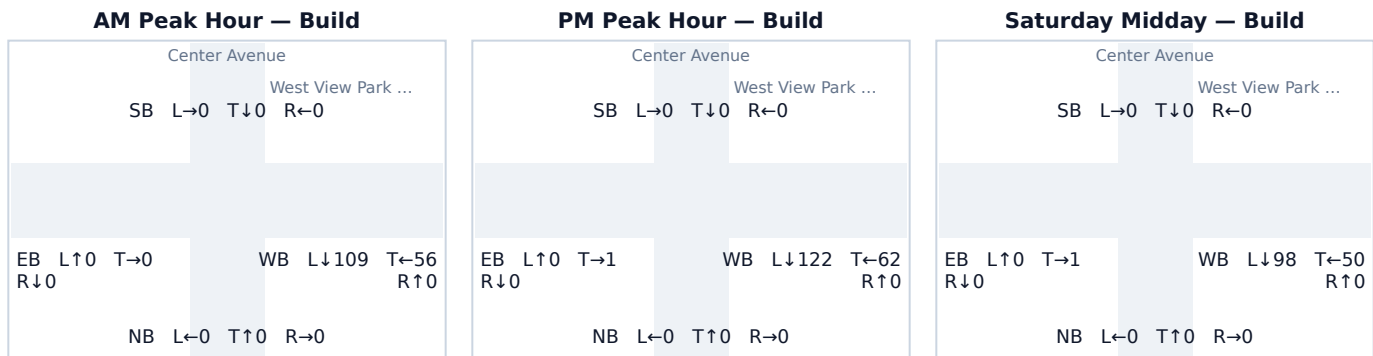
A.33 Center Avenue & West View Park Drive

Signal ID	pittsburgh-728
Location	NW Pittsburgh · 1.26 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS A · 5.1 s/veh · v/c 0.08
Build (PM)	LOS A · 5.1 s/veh · v/c 0.08
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 197% imbalance). 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0
EB approach	0	—	0	0
WB approach	118	60	—	178
Σ exiting	118	60	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	0	1	1	0.00 → 0.00	4.7 → 4.7	A → A	0
WB	183	1	184	0.17 → 0.17	5.6 → 5.6	A → A	56

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

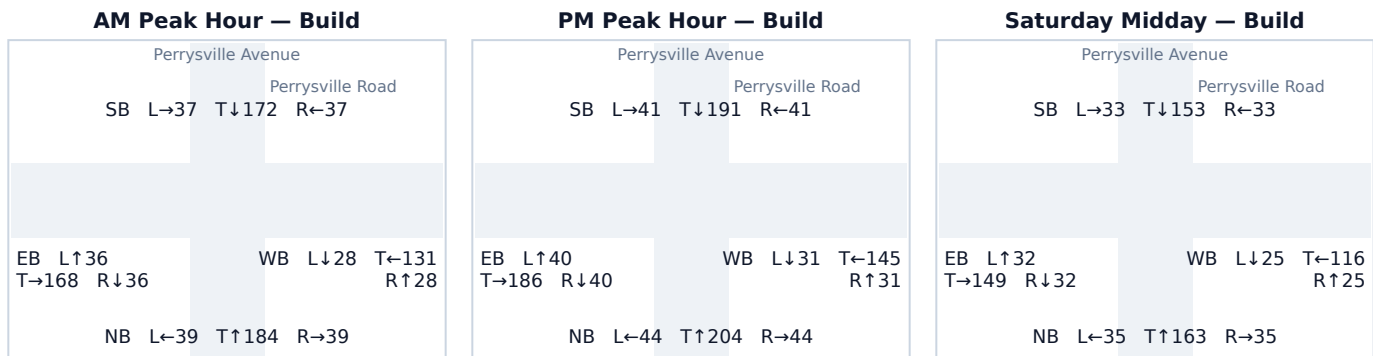
A.34 Perrysville Avenue & Perrysville Road

Signal ID	pittsburgh-2219
Location	NW Pittsburgh · 1.46 mi from site
Added PM peak trips	1
Existing / No-Build (PM)	LOS B · 16.2 s/veh · v/c 0.60
Build (PM)	LOS B · 16.2 s/veh · v/c 0.60
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.44 / EW 0.40; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: screening allocation (this junction did not resolve to the road network); turning movements: screening 15/70/15.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	292	0	292	0.37 → 0.37	12.8 → 12.8	B → B	135
SB	273	1	273	0.35 → 0.35	12.5 → 12.5	B → B	125
EB	266	0	266	0.37 → 0.37	14.2 → 14.3	B → B	129
WB	207	0	207	0.29 → 0.29	13.3 → 13.3	B → B	97

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	100%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.35 Perry Highway & Good Lane

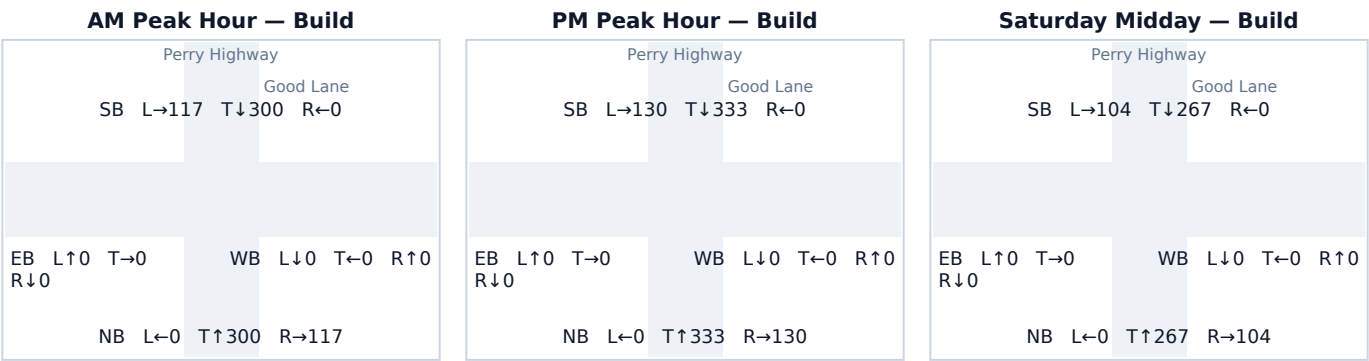
Signal ID	pittsburgh-2226
Location	NW Pittsburgh · 1.56 mi from site
Added PM peak trips	< 1 (negligible)
Existing / No-Build (PM)	LOS C · 21.6 s/veh · v/c 0.66
Build (PM)	LOS C · 21.6 s/veh · v/c 0.66
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 3 critical phases; g/C NS 0.35 / EW 0.23 (NS left 0.17); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 39% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

The development distributes fewer than one net PM peak car trip to this junction, so the Existing (No-Build) and Build conditions are numerically identical at reporting precision. The junction is reproduced here for completeness; the scheme's net car-mode trip generation is below the level at which junction capacity governs.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ WB leg	Σ entering
NB approach	—	324	126	450
SB approach	324	—	126	450
WB approach	0	0	—	0
Σ exiting	324	324	252	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	463	0	463	0.73 → 0.73	24.2 → 24.2	C → C	278
SB	463	< 1	463	0.73 → 0.73	24.2 → 24.2	C → C	278
EB	0	< 1	0	0.00 → 0.00	17.7 → 17.7	B → B	0
WB	0	< 1	0	0.00 → 0.00	17.7 → 17.7	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

A.36 Perry Highway & Lindley Lane

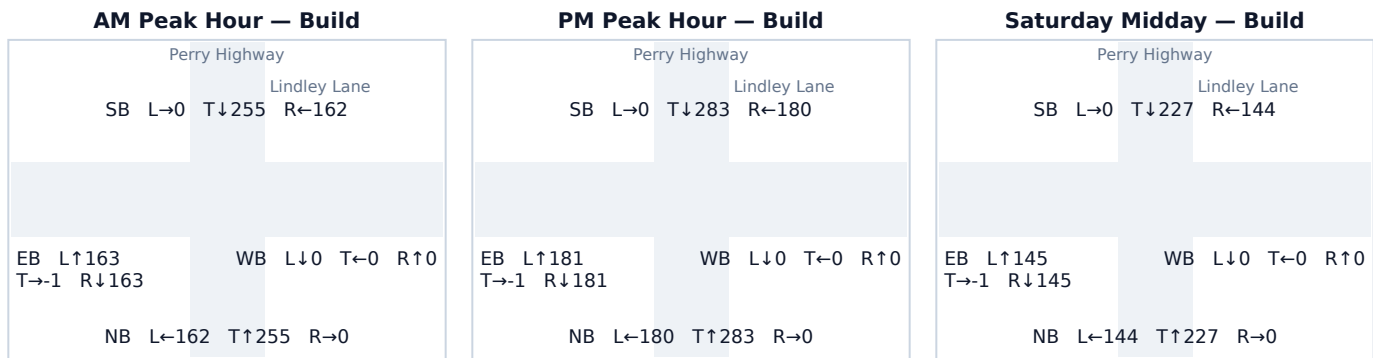
Signal ID	pittsburgh-2222
Location	NW Pittsburgh · 1.63 mi from site
Added PM peak trips	< 1 (negligible)
Existing / No-Build (PM)	LOS D · 42.9 s/veh · v/c 0.93
Build (PM)	LOS D · 42.9 s/veh · v/c 0.93
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +0.0s, design year +0.1s).

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 65 s cycle, 3 critical phases; g/C NS 0.35 / EW 0.27 (NS left 0.15); protected NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 4 iterations, residual 0.1 vph).

The development distributes fewer than one net PM peak car trip to this junction, so the Existing (No-Build) and Build conditions are numerically identical at reporting precision. The junction is reproduced here for completeness; the scheme's net car-mode trip generation is below the level at which junction capacity governs.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	Σ entering
NB approach	—	275	175	450
SB approach	275	—	175	450
EB approach	175	175	—	350
Σ exiting	450	450	350	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	463	0	463	0.74 → 0.74	26.6 → 26.6	C → C	304
SB	463	< 1	463	0.74 → 0.74	26.6 → 26.6	C → C	304
EB	361	0	361	0.74 → 0.74	31.6 → 31.6	C → C	245
WB	0	< 1	0	0.00 → 0.00	17.3 → 17.3	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.